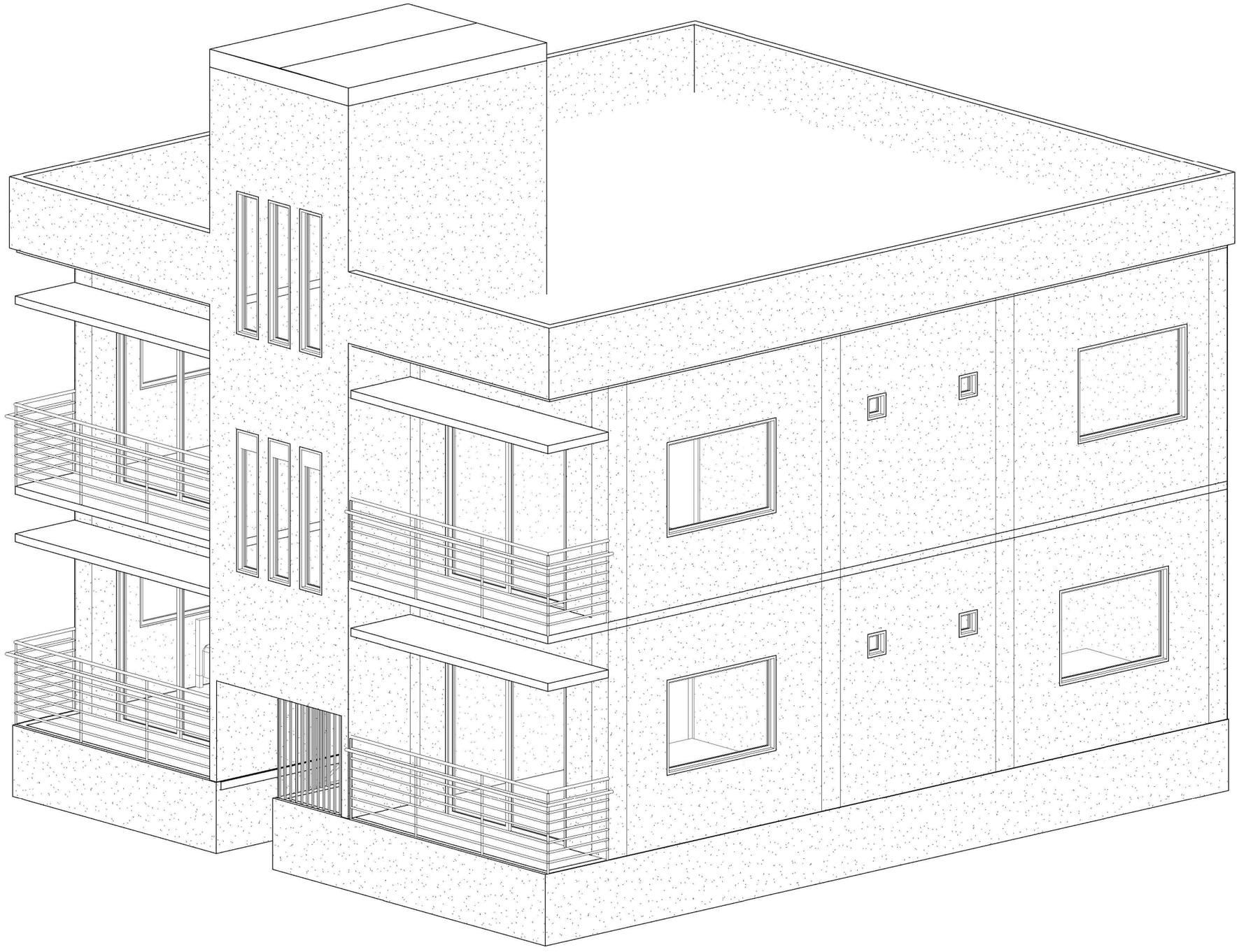
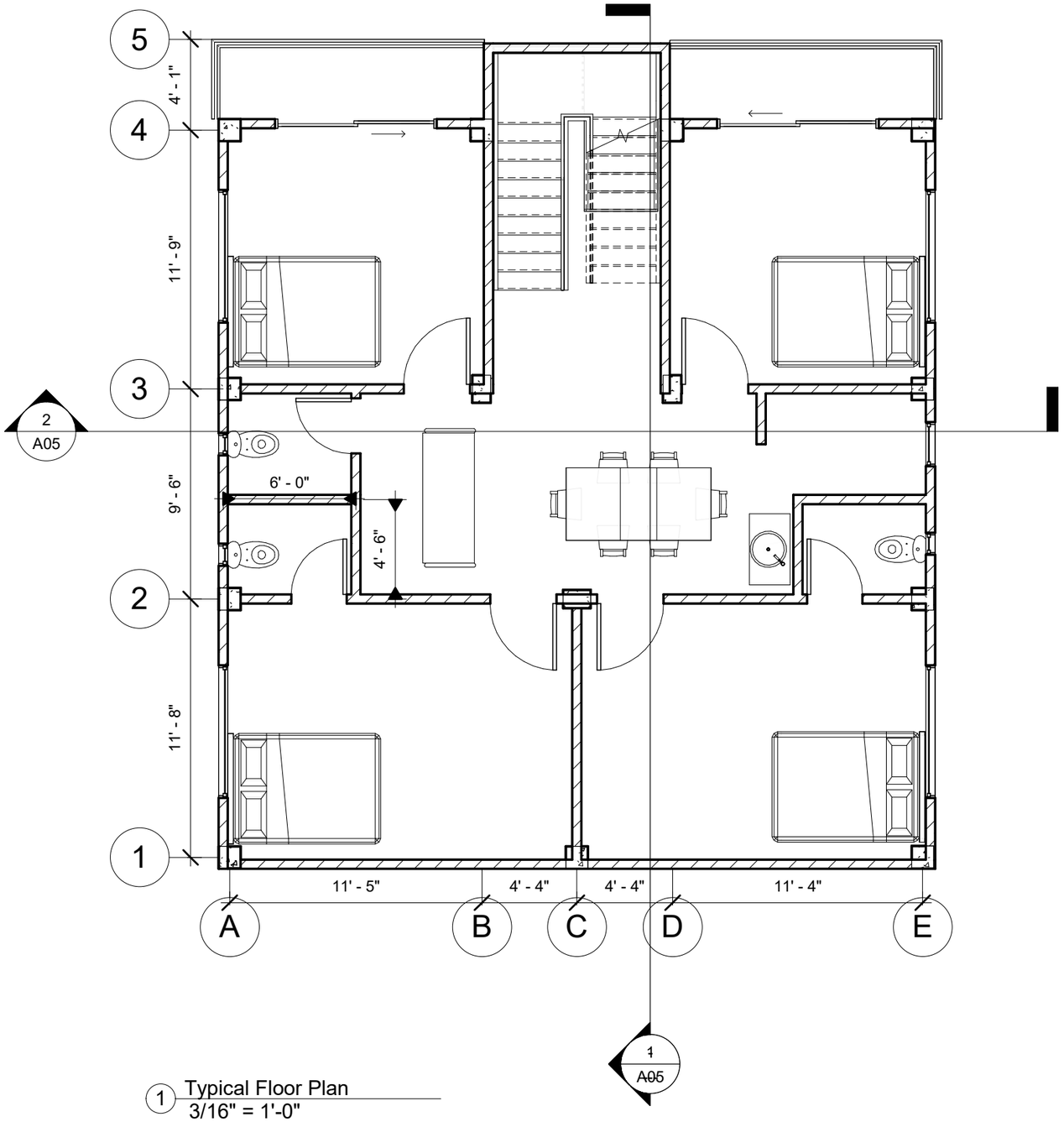
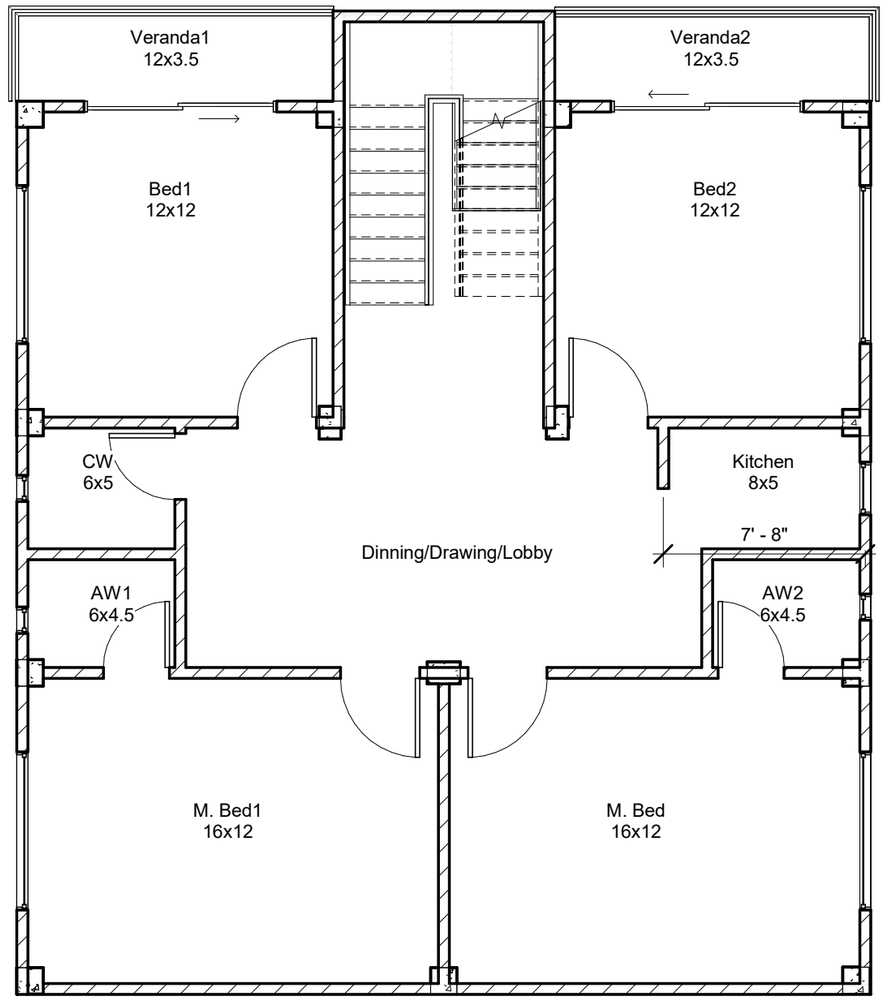


1 {3D}

Project	Authors	Sheet
N2 G+1 Story	Drawn By Shahlan Designed By Shahlan Kabir Checked By Shohel	3D A01 12/16/25



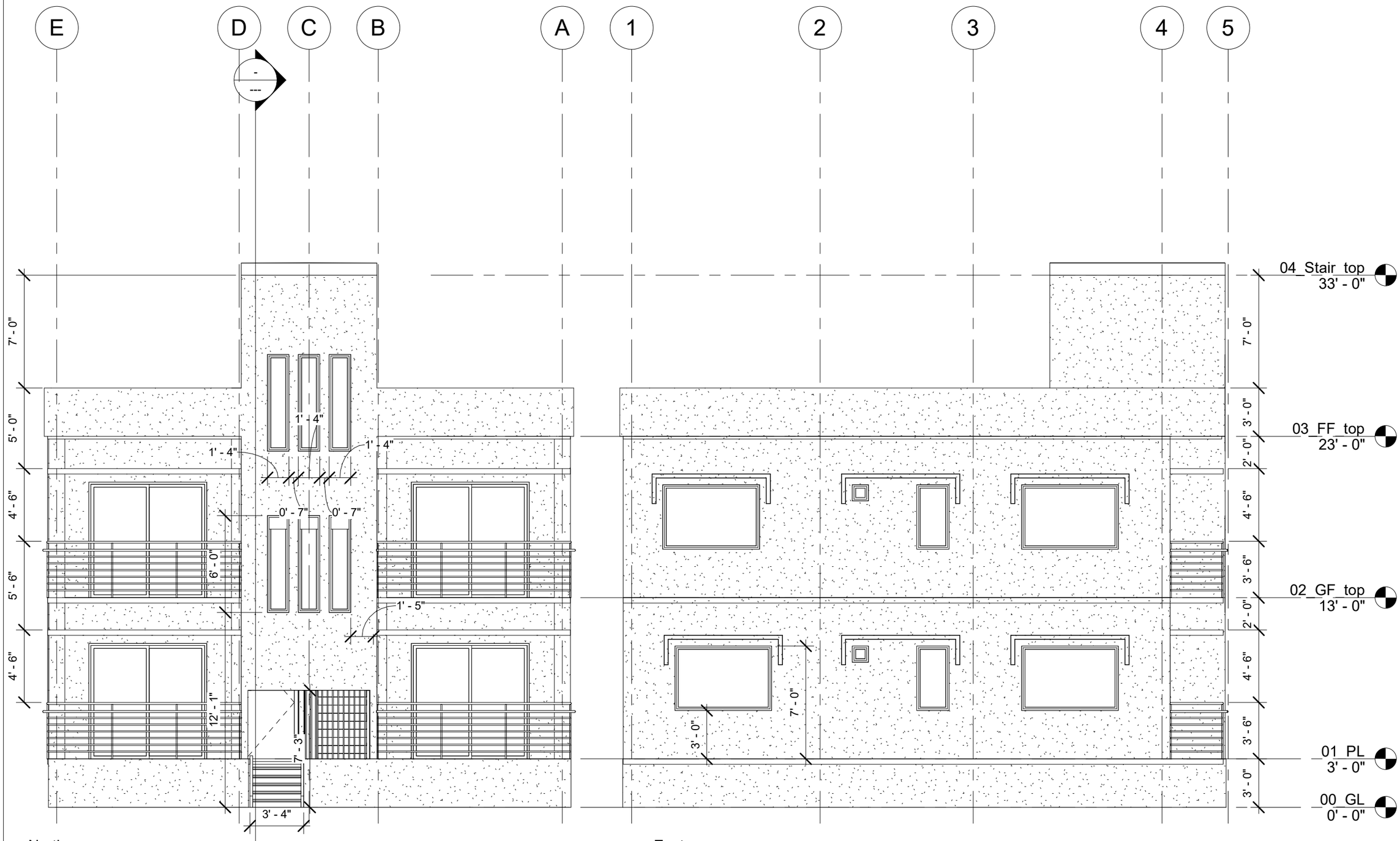
1 Typical Floor Plan
3/16" = 1'-0"



2 Typical Floor Rooms
3/16" = 1'-0"

1. Typical Floor Plan
2. Typical Floor Rooms

Project	Authors	Sheet
N2 G+1 Story	Drawn By Shahlan Kabir Designed By Shahlan Kabir Checked By Shohel	Floor Plan A02 12/16/25

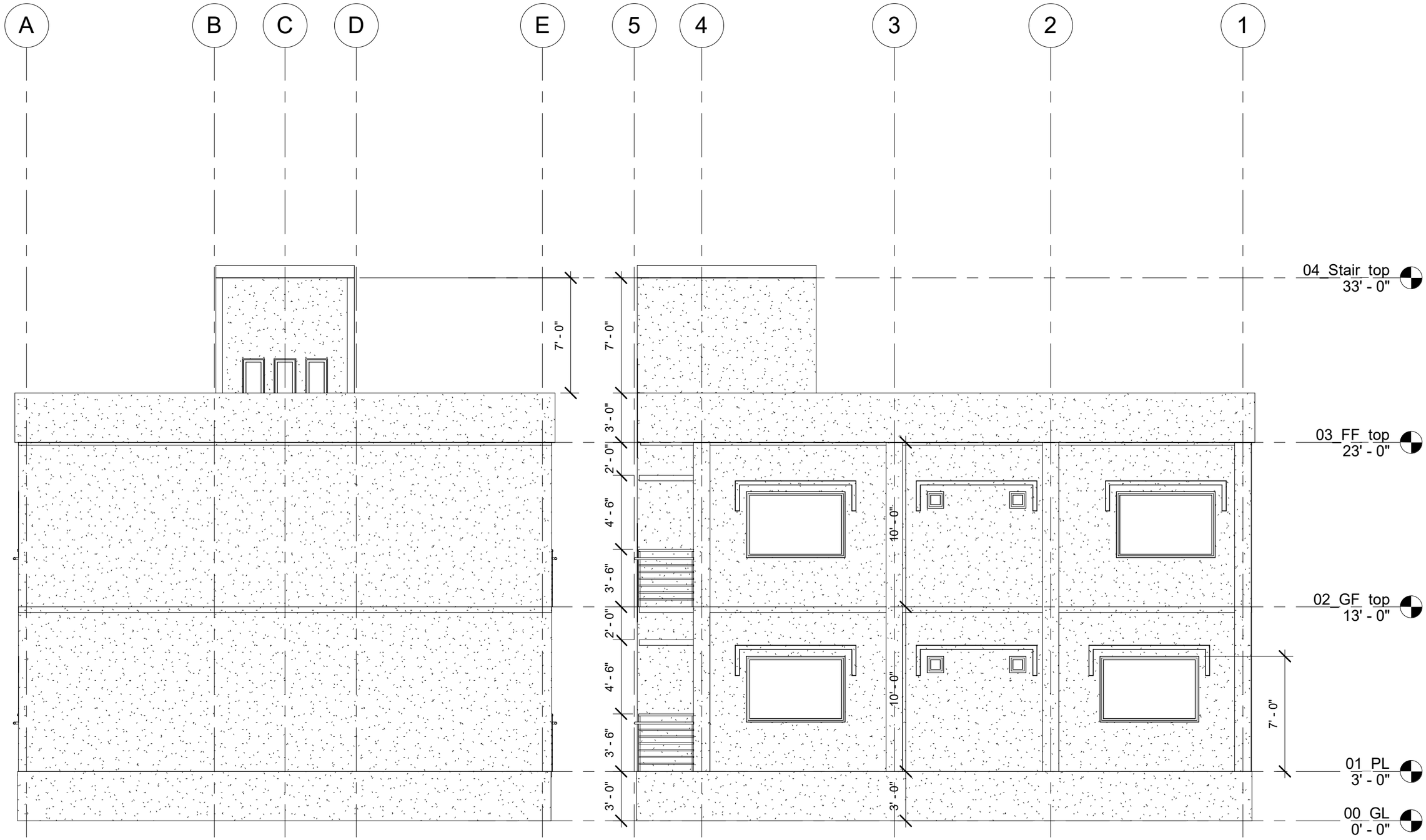


1 North
3/16" = 1'-0"

2 East
3/16" = 1'-0"

1. North Elevation
2. East Elevation

Project	Authors	Sheet
N2 G+1 Story	Drawn By Shahlan Kabir Designed By Shahlan Kabir Checked By Shohel	Elevation NE A03 12/16/25

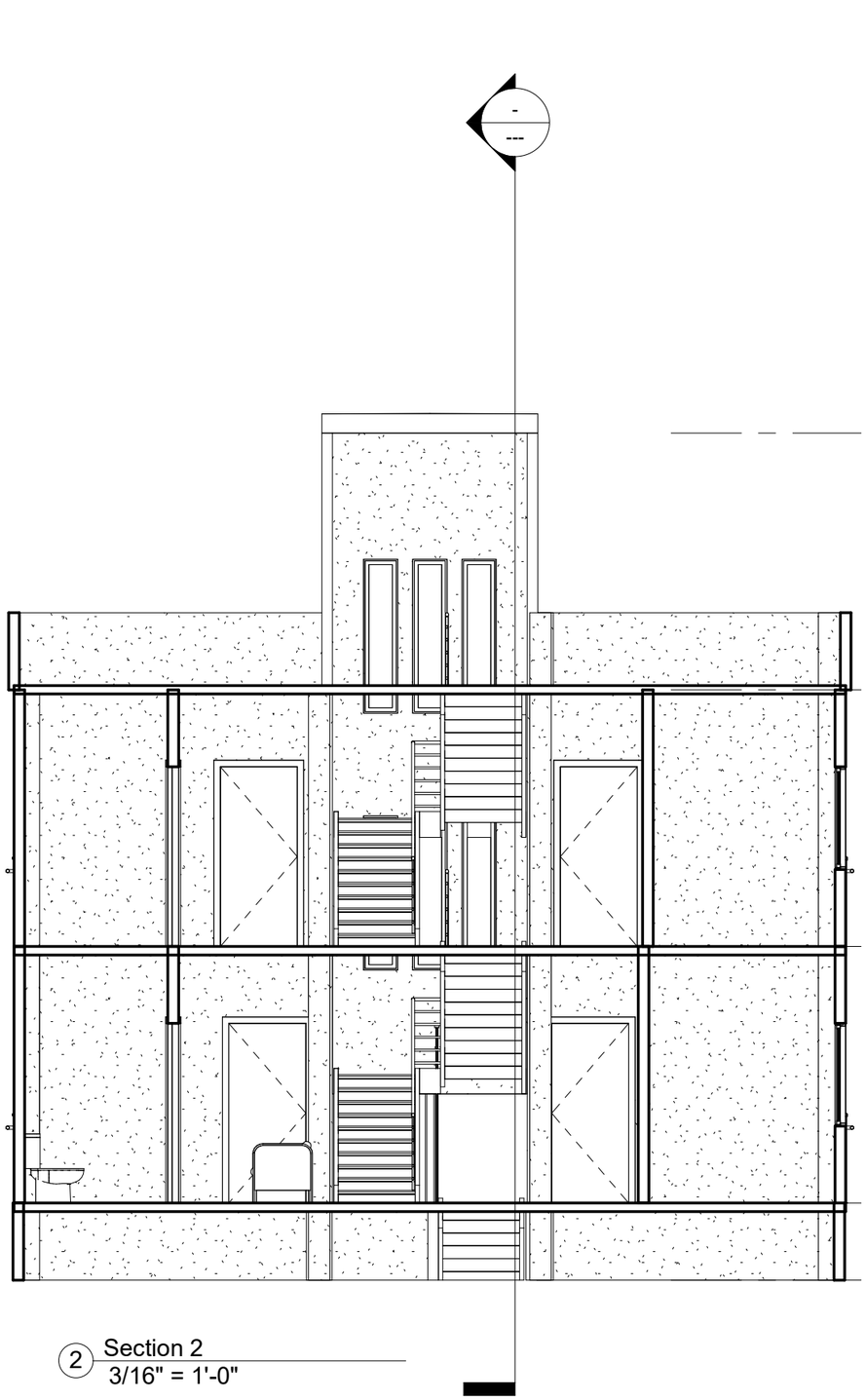


1 South
3/16" = 1'-0"

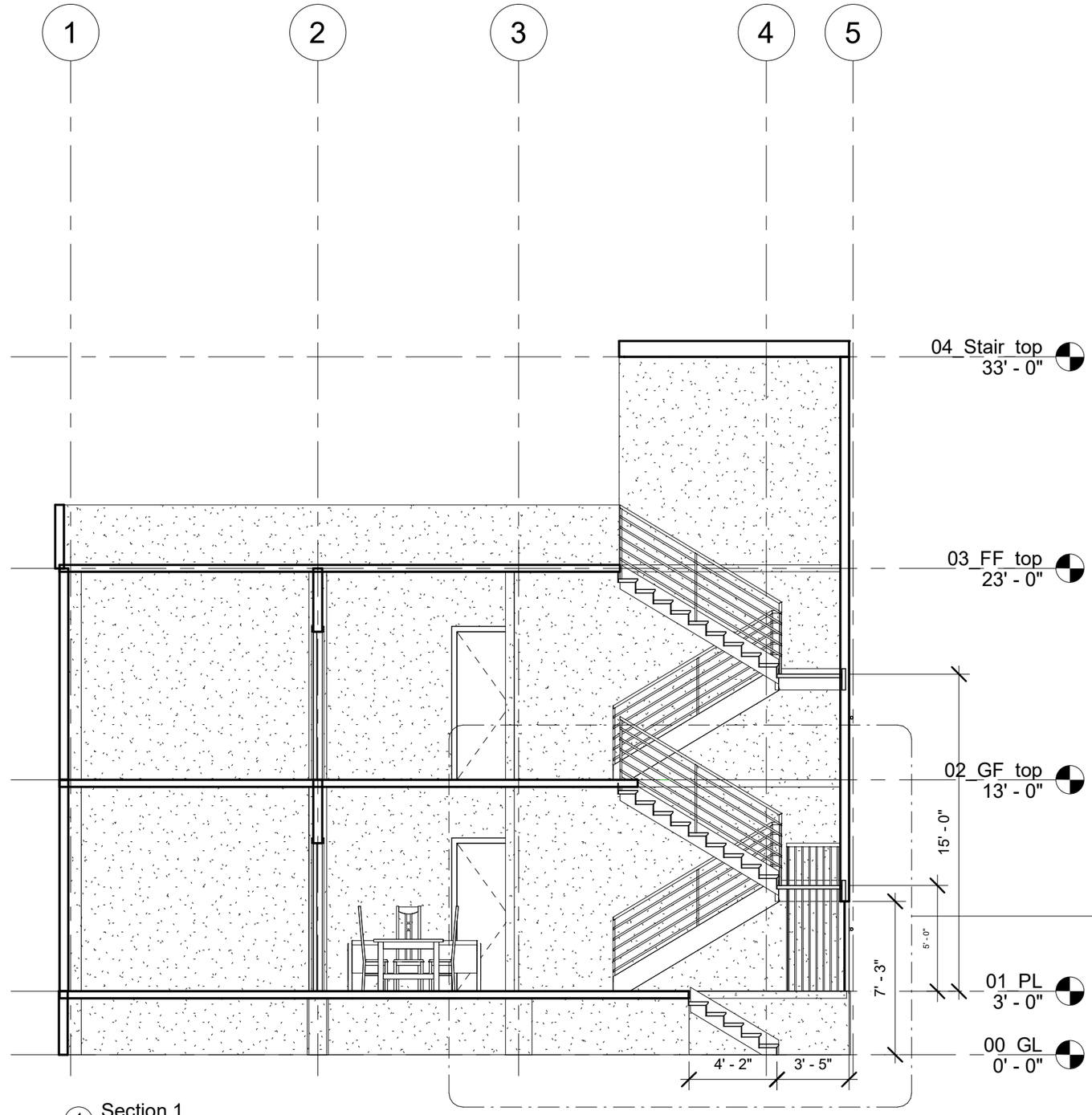
2 West
3/16" = 1'-0"

1. South Elevation
2. West Elevation

Project	Authors	Sheet
N2 G+1 Story	Drawn By Shahlan Kabir Designed By Shahlan Kabir Checked By Shohel	Elevation SW A04 12/16/25



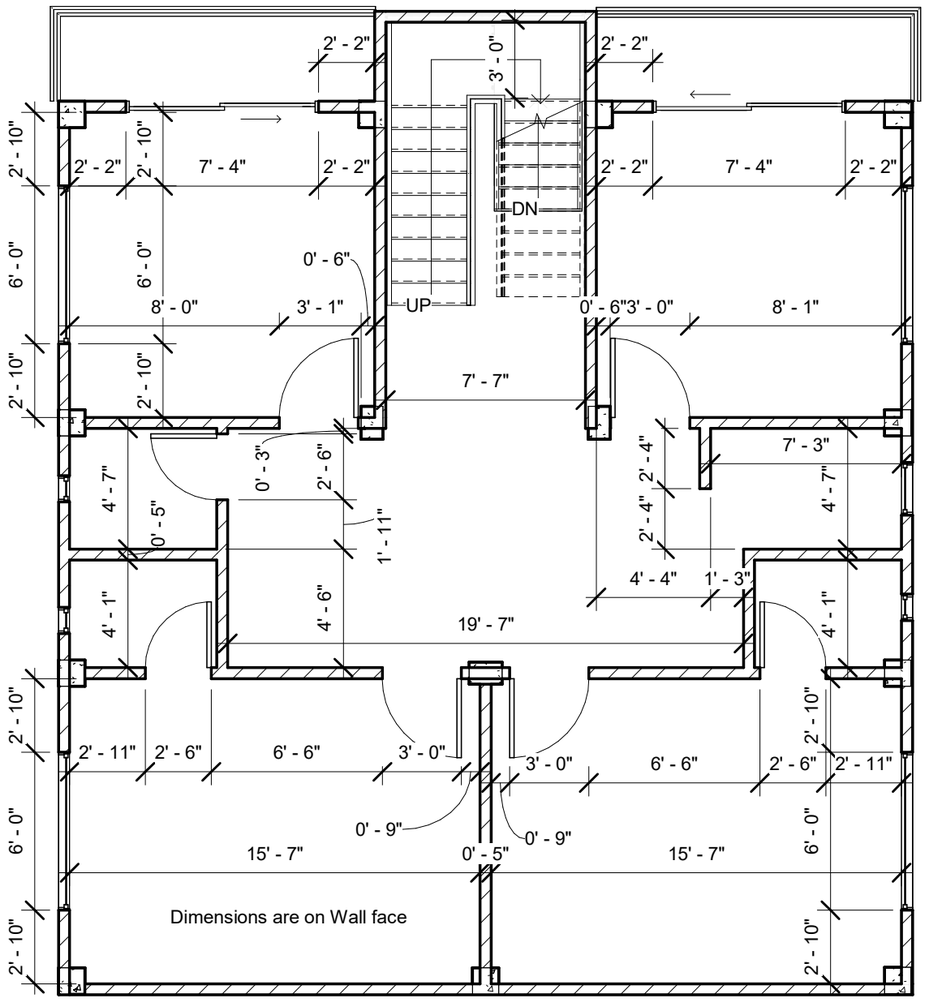
2 Section 2
3/16" = 1'-0"



1 Section 1
3/16" = 1'-0"

1. Section 1
2. Section 2

Project	Authors		Sheet
N2 G+1 Story	Drawn By	Shahlan Kabir	Sections
	Designed By	Shahlan Kabir	A05
	Checked By	Shohel	12/16/25



1 Wall Layout
3/16" = 1'-0"

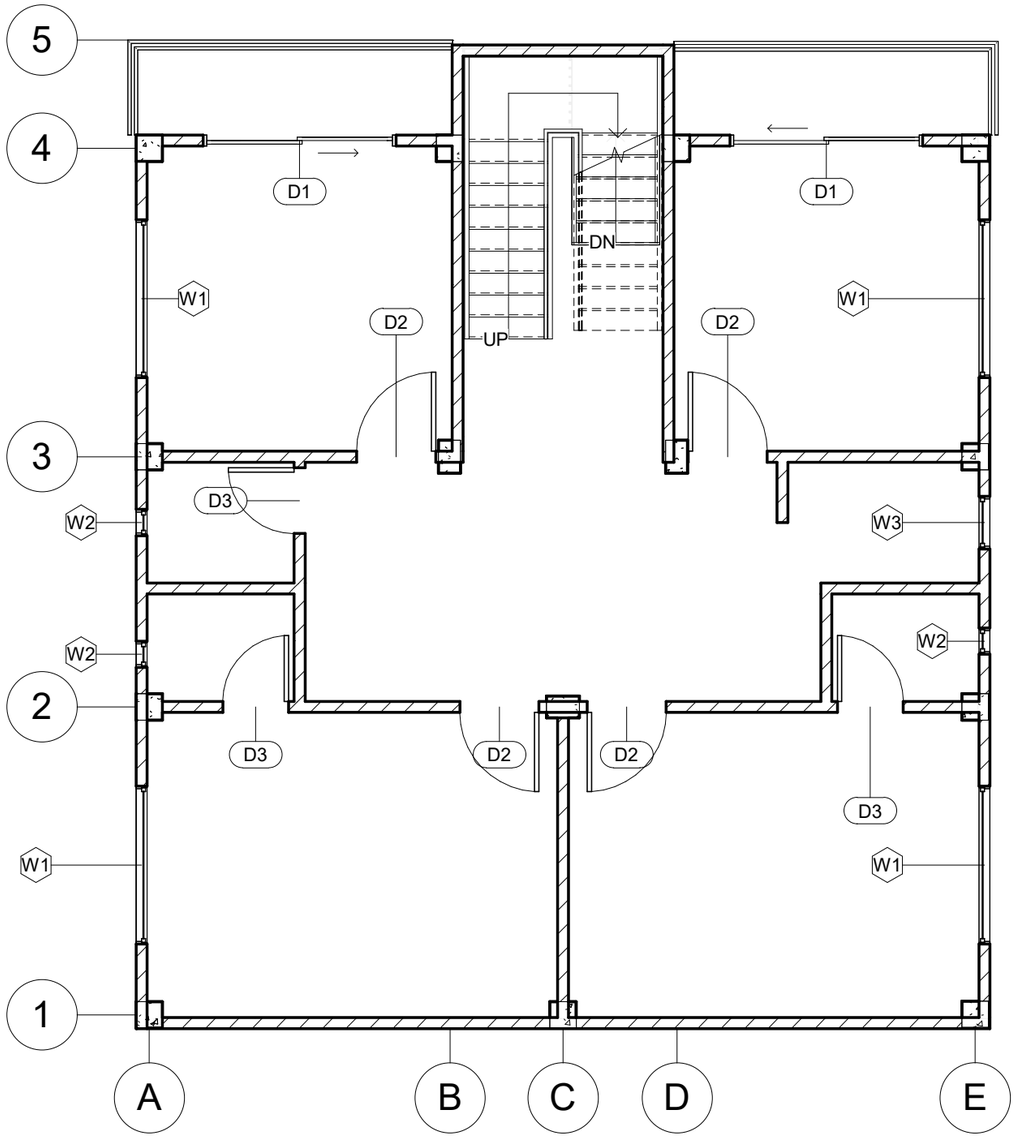
Wall Schedule		
Unconnected Height	Width	Length
10' - 0"	0' - 5"	2' - 6"
10' - 0"	0' - 5"	2' - 6"
10' - 0"	0' - 5"	4' - 6"
10' - 0"	0' - 5"	4' - 6"
10' - 0"	0' - 5"	4' - 6"
10' - 0"	0' - 5"	4' - 6"
10' - 0"	0' - 5"	5' - 0"
10' - 0"	0' - 5"	5' - 0"
10' - 0"	0' - 5"	6' - 0"
10' - 0"	0' - 5"	6' - 0"
10' - 0"	0' - 5"	6' - 0"
10' - 0"	0' - 5"	6' - 0"
3' - 0"	0' - 5"	7' - 4"
3' - 0"	0' - 5"	7' - 5"
25' - 9"	0' - 5"	8' - 0"
10' - 0"	0' - 5"	10' - 8"
10' - 0"	0' - 5"	10' - 8"
10' - 0"	0' - 5"	11' - 2 1/2"
3' - 0"	0' - 5"	12' - 0"
10' - 0"	0' - 5"	12' - 0"
10' - 0"	0' - 5"	12' - 0"
10' - 0"	0' - 5"	12' - 0"
10' - 0"	0' - 5"	12' - 0"
10' - 0"	0' - 5"	12' - 0"
10' - 0"	0' - 5"	12' - 0"
10' - 0"	0' - 5"	12' - 0"

Wall Schedule		
Unconnected Height	Width	Length
10' - 0"	0' - 5"	12' - 0"
10' - 0"	0' - 5"	12' - 0"
10' - 0"	0' - 5"	12' - 0"
10' - 0"	0' - 5"	15' - 5"
10' - 0"	0' - 5"	15' - 5"
10' - 0"	0' - 5"	15' - 5"
10' - 0"	0' - 5"	15' - 5"
3' - 0"	0' - 5"	16' - 2"
3' - 0"	0' - 5"	32' - 0"
10' - 0"	0' - 5"	32' - 0"
10' - 0"	0' - 5"	32' - 0"
10' - 0"	0' - 5"	32' - 0"
10' - 0"	0' - 5"	32' - 0"
3' - 0"	0' - 5"	32' - 5"
3' - 0"	0' - 5"	32' - 5"
10' - 0"	0' - 5"	32' - 8 1/2"
10' - 0"	0' - 5"	32' - 8 1/2"
10' - 0"	0' - 5"	33' - 6"
10' - 0"	0' - 5"	33' - 6"
3' - 0"	0' - 5"	37' - 0"
3' - 0"	0' - 5"	37' - 0"
3' - 0"	0' - 5"	37' - 1 1/2"
3' - 0"	0' - 5"	37' - 1 1/2"

Grand total: 48

1. Wall layout
2. Wall Schedule

Project	Authors	Sheet
N2 G+1 Story	Drawn By Shahlan Kabir Designed By Shahlan Kabir Checked By Shohel	Wall A06 01/20/26



1 Door & Window layout
3/16" = 1'-0"

Door Schedule			
Height	Sill Height	Width	Mark
7' - 0"	0' - 0"	3' - 0"	D2
7' - 0"	0' - 0"	3' - 0"	D2
7' - 0"	0' - 0"	3' - 0"	D2
7' - 0"	0' - 0"	3' - 0"	D2
7' - 0"	0' - 0"	2' - 6"	D3
7' - 0"	0' - 0"	2' - 6"	D3
7' - 0 3/8"	0' - 0"	7' - 0 21/32"	D1
7' - 0 3/8"	0' - 0"	7' - 0 21/32"	D1
7' - 0"	0' - 0"	2' - 6"	D3
7' - 0"	0' - 0"	3' - 0"	D2
7' - 0"	0' - 0"	3' - 0"	D2
7' - 0"	0' - 0"	3' - 0"	D2
7' - 0"	0' - 0"	2' - 6"	D3
7' - 0"	0' - 0"	2' - 6"	D3
7' - 0 3/8"	0' - 0"	7' - 0 21/32"	D1
7' - 0 3/8"	0' - 0"	7' - 0 21/32"	D1
7' - 0"	0' - 0"	2' - 6"	D3

Grand total: 18

Window Schedule			
Height	Width	Sill Height	Mark
4' - 0"	6' - 0"	3' - 0"	W1
4' - 0"	6' - 0"	3' - 0"	W1
4' - 0"	6' - 0"	3' - 0"	W1
4' - 0"	6' - 0"	3' - 0"	W1
1' - 0"	1' - 0"	6' - 0"	W2
1' - 0"	1' - 0"	6' - 0"	W2
1' - 0"	1' - 0"	6' - 0"	W2
4' - 0"	6' - 0"	3' - 0"	W1
4' - 0"	6' - 0"	3' - 0"	W1
4' - 0"	6' - 0"	3' - 0"	W1
4' - 0"	6' - 0"	3' - 0"	W1
1' - 0"	1' - 0"	6' - 0"	W2
1' - 0"	1' - 0"	6' - 0"	W2
1' - 0"	1' - 0"	6' - 0"	W2
6' - 0"	1' - 4"	9' - 1"	W4
6' - 0"	1' - 4"	9' - 1"	W4
6' - 0"	1' - 4"	9' - 1"	W4
6' - 0"	1' - 4"	9' - 1"	W4
6' - 0"	1' - 4"	9' - 1"	W4
6' - 0"	1' - 4"	9' - 1"	W4
4' - 0"	2' - 0"	3' - 0"	W3
4' - 0"	2' - 0"	3' - 0"	W3

Grand total: 22

1. Door and Window Layout
2. Door Schedule
3. Window Schedule

Project	Authors	Sheet
N2 G+1 Story	Drawn By Shahlan Kabir Designed By Shahlan Kabir Checked By Shohel	Door & Window A07 01/20/26

STRUCTURAL DESIGN CRITERIA & MINIMUM REQUIREMENTS FOR R.C.C.

1. GENERAL

a) DESIGN METHOD USED IS USD ACCORDING TO BANGLADESH NATIONAL BUILDING CODE (BNBC) 2020, UBC 1997 SUPPLEMENTED BY ACI 318-08.
b) ALL THE STRUCTURAL DRAWINGS SHALL BE READ IN CONJUNCTION WITH RELEVANT ARCHITECTURAL DRAWINGS.
c) FOLLOW BNBC-2020 FOR SPECIFICATIONS/STRUCTURAL REQUIREMENTS NOT MENTIONED IN THE DRAWINGS OR IN THIS NOTE SHEET.
d) ANY DETAILS NOT SHOWN IN THE DRAWING SHOULD BE DONE ACCORDING TO ACI DETAILING MANUAL.
e) BASIC WIND SPEED = 57.1 m/sec, EXPOSURE CATEGORY EXPOSURE-A STRUCTURE IMPORTANCE COEFFICIENT I, = 1.0
f) SEISMIC ZONE - 2, ZONE COEFFICIENT Z = 0.2, STRUCTURE IMPORTANCE COEFFICIENT I = 1.0
g) LIVE LOAD ON FLOOR & ROOF = 2 KN/sqm (NO REDUCTION FOR LIVE LOADS)
h) OTHER LOADS AS PER BNBC 2020.
i) SUPER STRUCTURE HAS BEEN DESIGNED AS INTERMEDIATE MOMENT RESISTING FRAME (IMRF) WITH NECESSARY JOINT DETAILING AND DUCTILITY REQUIREMENT FOR SEISMIC ZONE - 2

2. FOUNDATION

a)FOUNDATION HAS BEEN DESIGNED FOR SAL FOREST HOUSE
b) FOUNDATION TYPE - FOOTING FOUNDATION HAS BEEN DESIGNED ACCORDING TO THE SOIL TEST REPORT.

3. CONCRETE

a) TYPE : CONCRETE SPECIFIED COMPRESSIVE STRENGTH CONSIDERED AS FOLLOWS :
FOR 28 DAYS CONCRETE CYLINDER STRENGTH
i) GRADE BEAM,SLAB, BEAM, f'c=min.21MPa (3,000psi).
b) MINIMUM CYLINDER STRENGTH :
BASED ON CYLINDER TEST OF DIAMETER D = 150mm & HEIGHT = 300mm
i) 28 DAYS STRENGTH = 100% OF REQUIRED AVERAGE COMPRESSIVE STRENGTH
ii) 7 DAYS STRENGTH = 70% OF REQUIRED AVERAGE COMPRESSIVE STRENGTH
c) METHOD OF CURING :
* HORIZONTAL SURFACE - BY POUNDING OF WATER
* OTHER SURFACES -BY WRAPPING MOIST JUTE FABRIC AND SPARKLING WATER BY HOSE PIPE FREQUENTLY.

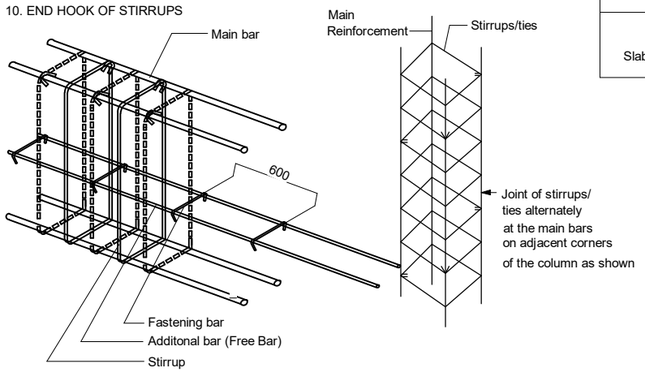
4. CEMENT

PORTLAND COMPOSITE CEMENT TYPE -2, ASTM-C 150, BDS EN-197-1-CEM II (42.5 N)

5. CONCRETE AGGREGATE

a) FINE AGGREGATES : SYLHET SAND WITH F.M > 2.5 (Minimum)
b) COARSE AGGREGATES : 20mm (4") DOWN GRADE CRUSHED STONE FROM BOULDERS TO BE USED AS COARSE AGGREGATE IN ALL R.C.C. WORK EXCEPT RAILING, DROP WALL, FINS FOR WHICH 12mm DOWN GRADED STONE CHIPS TO USED.

10. END HOOK OF STIRRUPS



END HOOKS OF STIRRUPS ARE LOCATED ALTERNATELY AT TOP CORNER BAR OF THE SECTION.

6. WATER

POTABLE WATER TO BE USED IN ALL CONCRETE MIX.

7. STEEL REINFORCEMENT

a) ALL REINFORCEMENTS ARE OF = 500 MPa (72.5 GRADE) DEFORMED BAR CONFORMING TO ONE OF THE FOLLOWING SPECIFICATIONS :
i) BDS ISO 6935-2:2006
ii) ASTM A615M A616M A617M
b) THE FOLLOWING TESTS FOR REINFORCING BARS FROM RANDOM SAMPLES SHALL BE CONDUCTED AT BUET / APPROVED DEPARTMENT AS PER BDS ISO 6935-2:2006 AND TEST RESULT SHALL BE SUBMITTED TO THE ENGINEER FOR CHECKING & RECORD :
i) TENSILE STRENGTH TEST
ii) PERCENTAGE ELONGATION TEST
iii) BEND TEST

8. LAP LENGTH

UNLESS OTHERWISE MENTIONED IN THE DRAWINGS, LAP LENGTH OF BARS SHALL BE :

f 'y=420 MPa & f 'c=25 MPf 'y=500 MPa & f 'c=21 MPa

BAR DIA (mm)	TOP BARS (mm)	BOTTOM BARS (mm)	TOP BARS (mm)	BOTTOM BARS (mm)	COMPRESSION (mm)
10∅	525	425	589	454	300
12∅	650	500	707	544	375
16∅	850	650	943	725	475
20∅	1075	825	1179	907	600
22∅	1450	1125	1621	1247	675
25∅	1650	1275	1842	1417	750

COLUMN SHALL BE TENSION LAPS OF TOP BARS

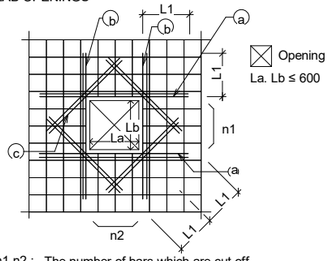
WELDED LAP SHALL BE PERMITTED IF EACH BAR IS MARKED WITH " W " ON THE BODY OF THE BAR AT EVERY METER AND CENTER TO CENTER DISTANCE BETWEEN THE SPLICES SHALL BE MINIMUM 600mm.
ONLY 50% OF THE BAR SHALL BE LAPPED AT ONE PLACE.

9. RECOMMENDED END HOOKS :

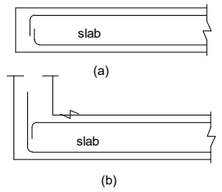
Bar Hook	Form and Extention	Bend Angle (degree)	Bend Radius (r)	Used Bar	Location
Primary reinf.	db, 4db	180°	r = 2db	6 to 25	lap splice end of anchorage
Tie / Stirrups	db, 6db	135°	r = 1.25db	6 to 12	stirrups fastening bar diagonal hoop
Primary reinf.	db, 2db	90°	r = 2.5db	6 to 32	bend up for embedment
Slab	db	45°	r = 5db	6 to 20	bend bar in slab

10a. REINFORCEMENT DETAILS

FOR SLAB OPENINGS

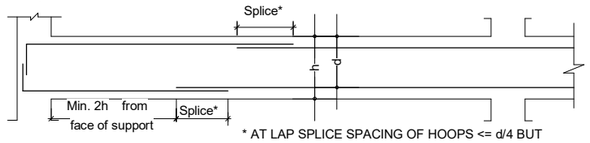


n1,n2 : The number of bars which are cut off for opening
(a) The number of extra bars > n1/2 (top and bott bars)
(b) The number of extra bars > n2/2 (top and bott bars)



11. STANDARD HOOKS OF REBAR
FOR ALL RE-BAR : PROVIDE 90° STANDARD HOOKS (L-BENT) PLUS 12 d EXTENSION AT END SUPPORT/FREE END, IF NOT SHOWN IN THE DRAWINGS.
12. SPACER BARS
TO SUPPORT SECOND LAYER BARS IN BEAMS USE 25∅ SPACER BARS @ 2100mm C/C WHERE REQUIRED IF NOT SHOWN IN THE DRAWINGS.
13. CHAIRS b
USE CHAIRS OF NECESSARY DIMENSION MADE OF 10∅ BAR TOP SUPPORT TOP BARS OF SLAB @ 2000 C/C.

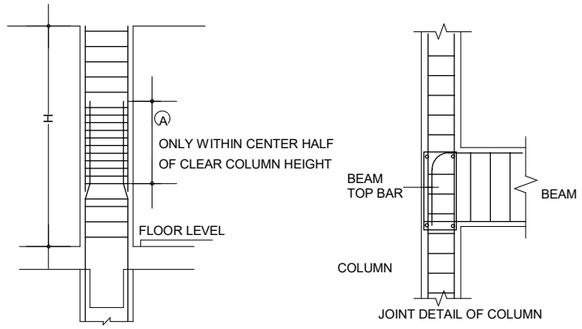
14. LAP LOCATION
a) FOR BEAM BOTTOM BAR, LAP NOT TO BE PROVIDED AT MIDDLE THIRD ZONE OF THE SPAN.
b) FOR BEAM TOP BAR, LAP MAY BE PROVIDED AT MIDDLE THIRD ZONE OF THE SPAN. NOT MORE THAN 50% OF THE BARS SHALL BE
c) SPLICED AT ONE PLACE.
d) LAP SPLICES ARE TO BE CONFINED BY HOOKS WITH MAXIMUM SPACING OR PITCH OF d/4 OR 100mm WHICHEVER IS SMALLER.
e) BEAM SPLICED LOCATION.VE DEPTH OF THE BEAM.



* AT LAP SPLICE SPACING OF HOOPS <= d/4 BUT NOT GREATER THAN 100mm.

15. MAXIMUM BARS IN BEAMS IN SINGLE LAYER
MAXIMUM NUMBER OF BARS AS A SINGLE LAYER IN BEAM STEM SHALL BE AS PER ACI DETAILING MANUAL.

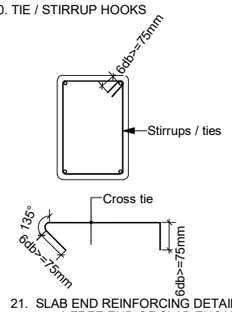
f) COLUMN SPLICE LOCATION



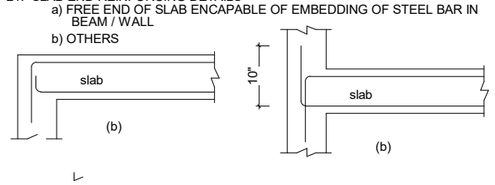
(A) IS AREA FOR SPLICE OF COLUMN REINFORCEMENT. MAXIMUM 50% OF TOTAL BAR IS SPLICED AT ONE LEVEL. H = CLEAR COLUMN HEIGHT

16. MINIMUM BAR SPACING OF COLUMN LONGITUDINAL BARS
CLEAR DISTANCE BETWEEN LONGITUDINAL BARS SHALL NOT BE LESS THAN 1.5 TIMES BAR DIAMETER 1.5 TIMES OF THE MAXIMUM SIZE OF COARSE AGGREGATE NOR 40mm.

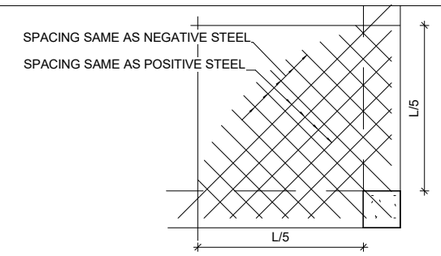
20. TIE / STIRRUP HOOKS



21. SLAB END REINFORCING DETAILS
a) FREE END OF SLAB ENCAPABLE OF EMBEDDING OF STEEL BAR IN BEAM / WALL
b) OTHERS

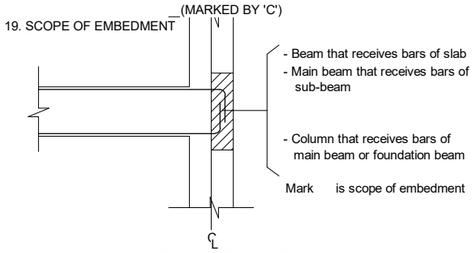


17. DEVELOPMENT DEPTH
ALL BEAM AND SLAB RE BARS SHOULD BE EXTENDED INTO THE SUPPORT UP TO DEVELOPMENT DEPTH.
18. CORNER REINFORCEMENT ('C')
CORNER REINFORCEMENT FOR BEAM SUPPORTED 2-WAY SLABS.



DETAIL OF CORNER REINF. (MARKED BY 'C')

19. SCOPE OF EMBEDMENT

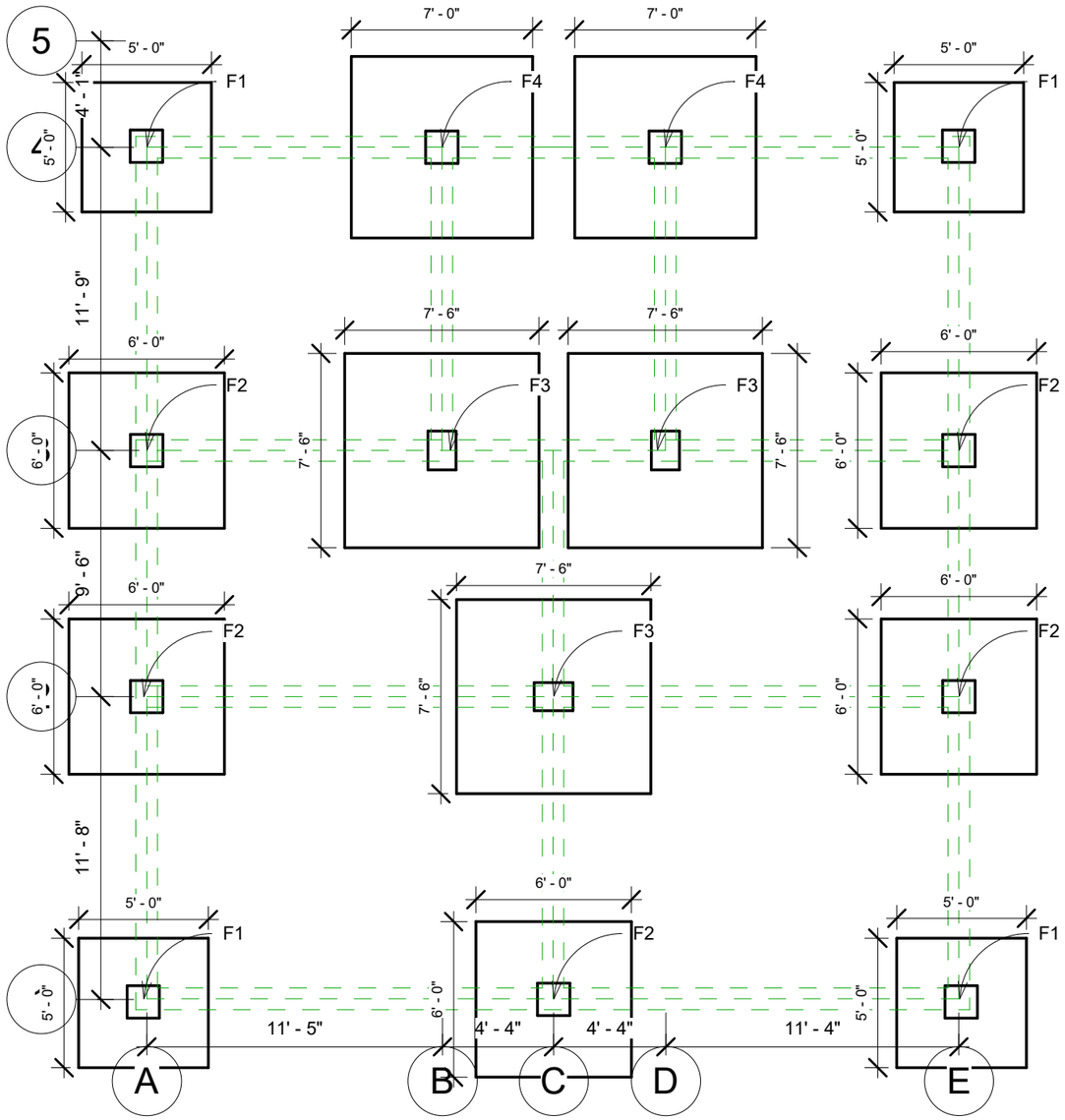


22. CONCRETE CLEAR COVER

Member	Location/Condition	Clear Cover(mm)	Figure
PILE CAP	Side	3"	3" 3"
	Bottom	3"	
Column	Above G.L	* 1.5"	1.5" 1.5" 1.5" 1.5"
	Below G.L	* 3"	3" 3" 3" 3"
Wall	Above G.L	* 1.5"	1.5" 1.5"
	Below G.L	* 3"	3" 3"
Retaining Wall	Exterior	3"	3" 1.5"
	Interior	1.5	
Beam	Top, Side	** 1.5"	1.5"
	Bottom	** 1.5"	1.5" 1.5"
Grade Beam	Top, Side	** 3"	3" 3"
	Bottom	** 3"	3" 3"
Slab and Stair	Top	.75"	.75"
	Bottom	1"	1"
Water Tank	Water face	2"	3" 2"
	SOIL face	3"	2" 3"

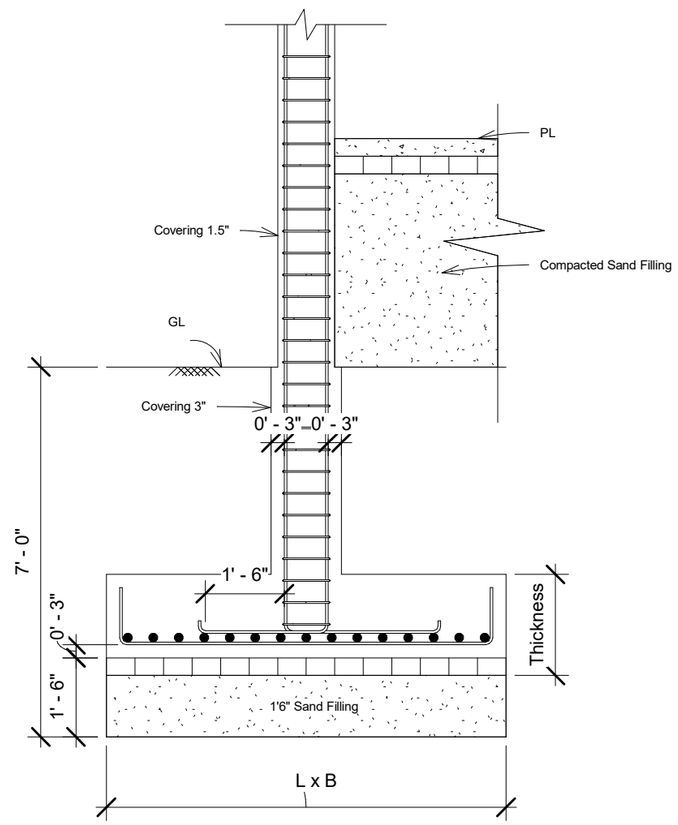
* From tie ** From stirrups

	Project	Authors	Sheet
General Guide lines	N2 G+1 Story	Drawn By Shahlan Kabir Designed By Shahlan Kabir Checked By Seikh Arif Istiak	General Guidelines S00 02/07/26



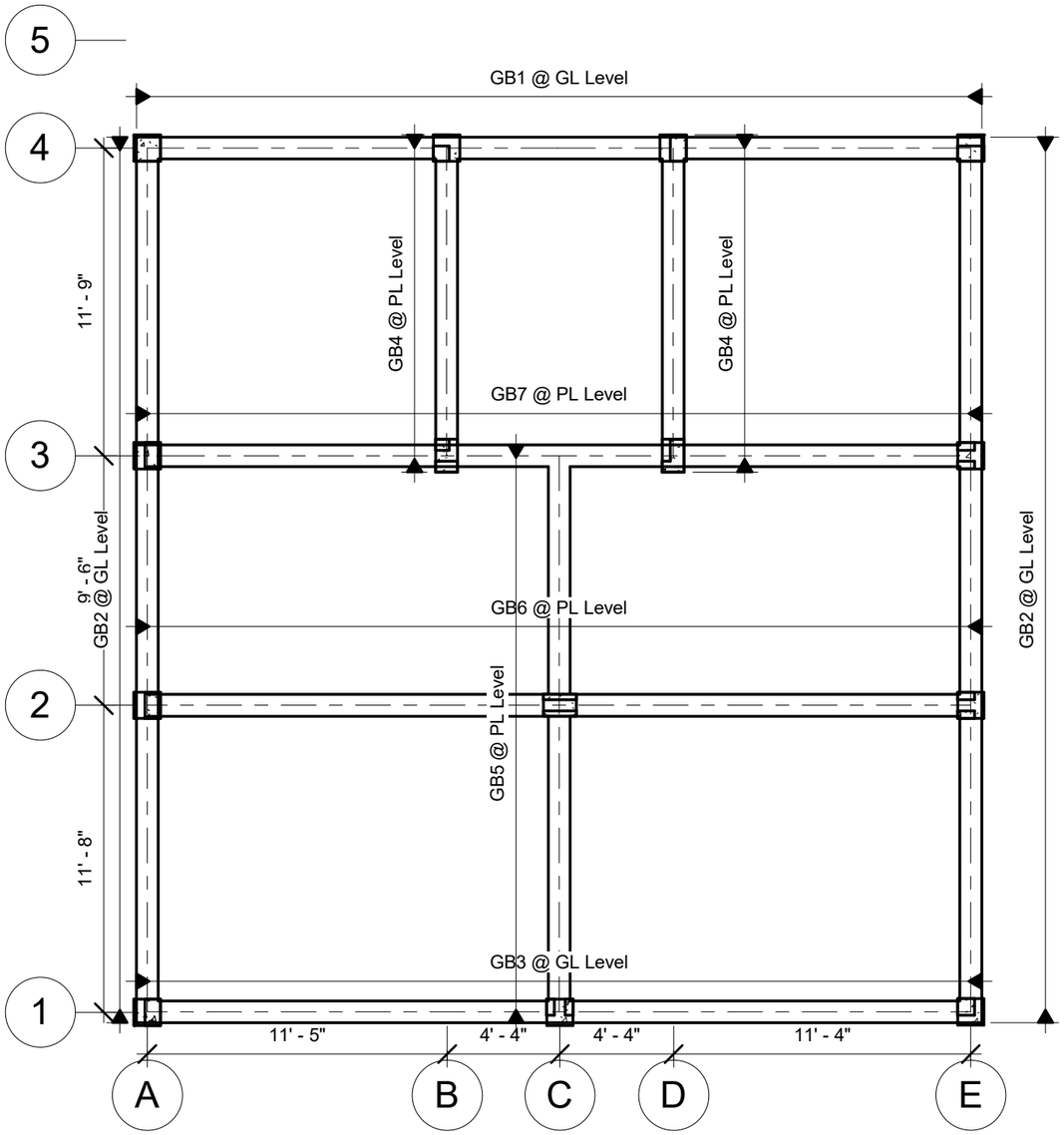
1 Footing Layout
3/16" = 1'-0"

Footing Schedule						
Foundation	Number of Footing	Loads associate (kip)...	Size of Footing L x L	Reinforcement in 'A' Direction	Reinforcement in 'B' Direction	Thickness
F1	4	27;8	5' x 5'	7-12mm bar dia	7-12mm bar dia	13"
F2	5	43;12	6' x 6'	10-12mm bar dia	10-12mm bar dia	15"
F3	3	66;26	7.5' x 7.5'	11-12mm bar dia	11-12mm bar dia	15"
F4	2	58;27	7' x 7'	11-12mm bar dia	11-12mm bar dia	15"



2 Footing section
3/8" = 1'-0"

Project	Authors	Sheet
N2 G+1 Story	Drawn By Shahlan Kabir Designed By Shahlan Kabir Checked By Seikh Arif Istiak	Foundation Layout S01 02/07/26



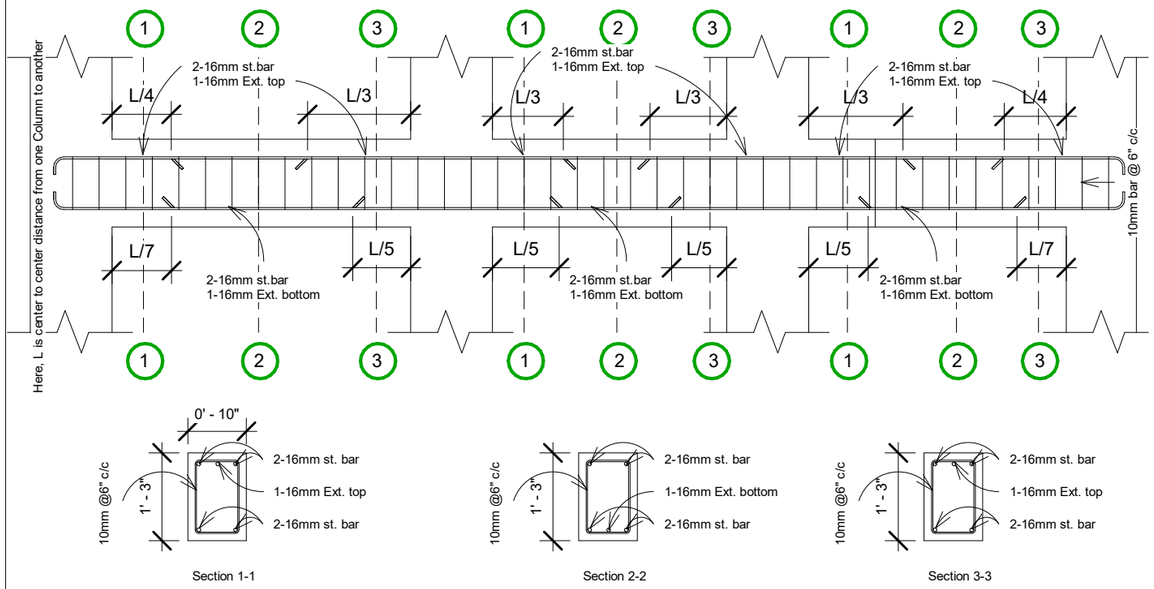
Grade Beam Grid Location			
Tag	**Elv**	**Dim**	**Grid**
GB1	GL	10" x 15"	4
GB2	GL	10" x 15"	A
GB2	GL	10" x 15"	E
GB3	GL	10" x 15"	1
GB4	PL	10" x 15"	B
GB4	PL	10" x 15"	D
GB5	PL	10" x 15"	C
GB6	PL	10" x 15"	2

1 Grade Beam layout
3/16" = 1'-0"

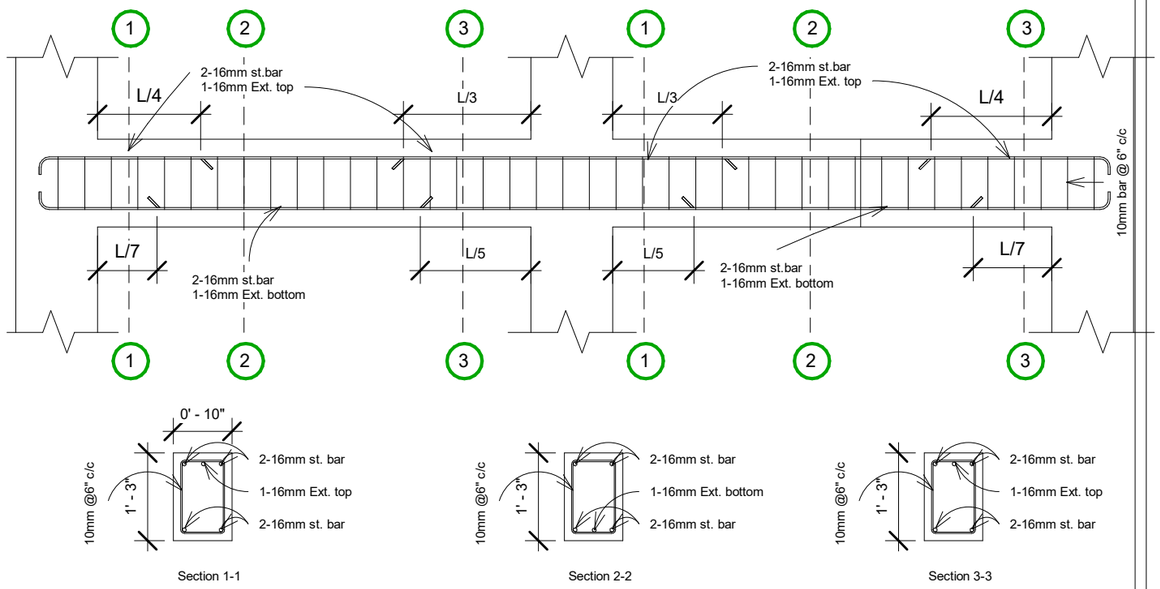
1. Grade Beam Layout
2. Grame Beam Grid Location

Project	Authors	Sheet
N2 G+1 Story	Drawn By Shahlan Kabir Designed By Shahlan Kabir Checked By Seikh Arif Istiak	Grade Beam Layout S02 01/22/26

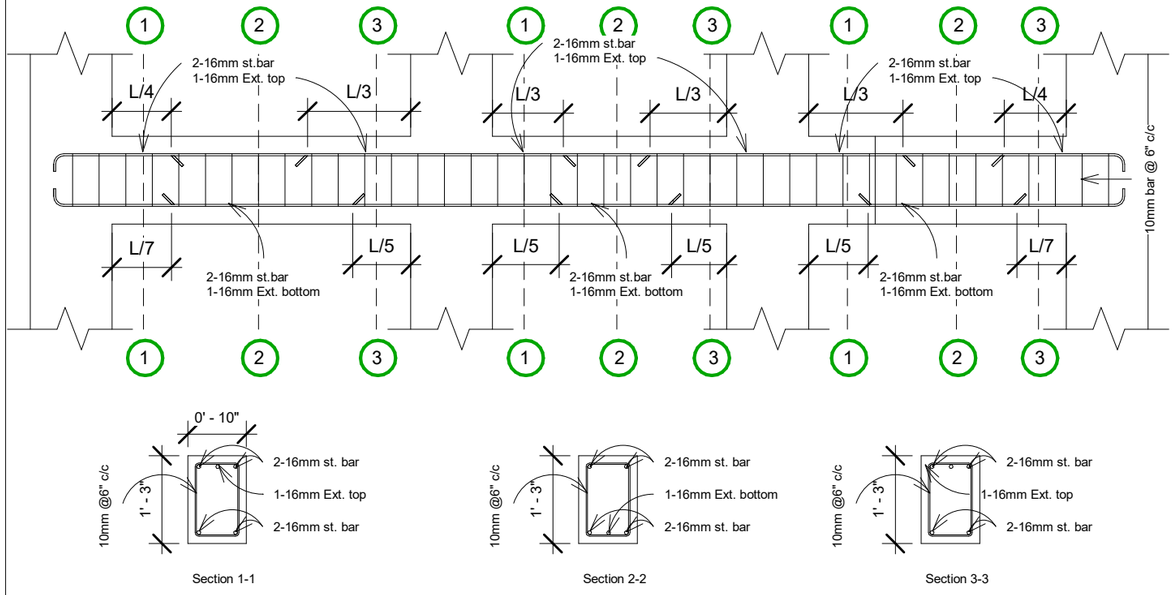
GB1 @ GL Level



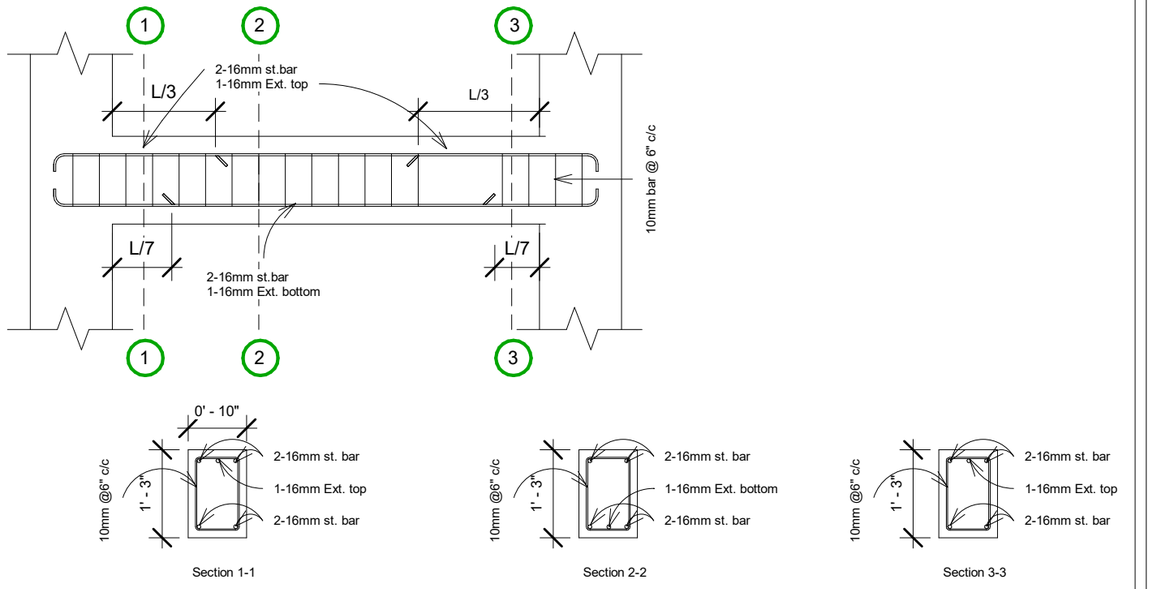
GB3 @ GL Level



GB2 @ GL Level



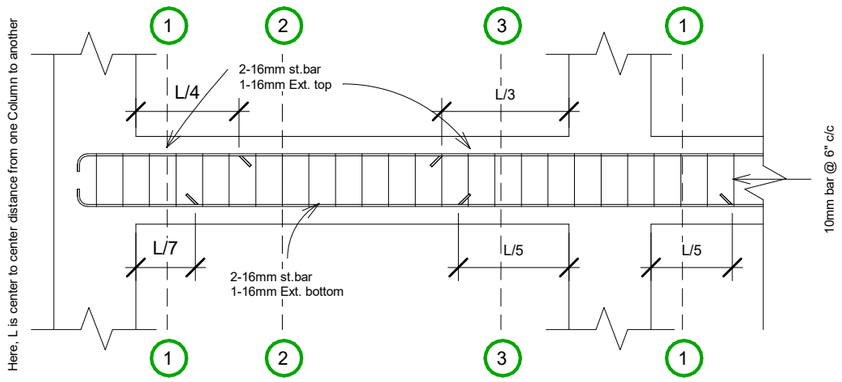
GB4 @ PL Level

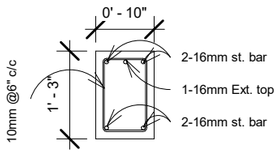


1. Grade Beam Schedule Part 01
* Not to Scale

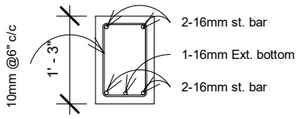
Project	Authors	Sheet
N2 G+1 Story	Drawn By Shahlan Kabir Designed By Shahlan Kabir Checked By Seikh Arif Istiak	Grade Beam Schedule 01 S03 02/06/26

GB5 @ PL Level

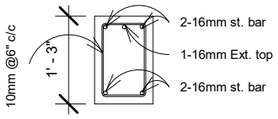




Section 1-1

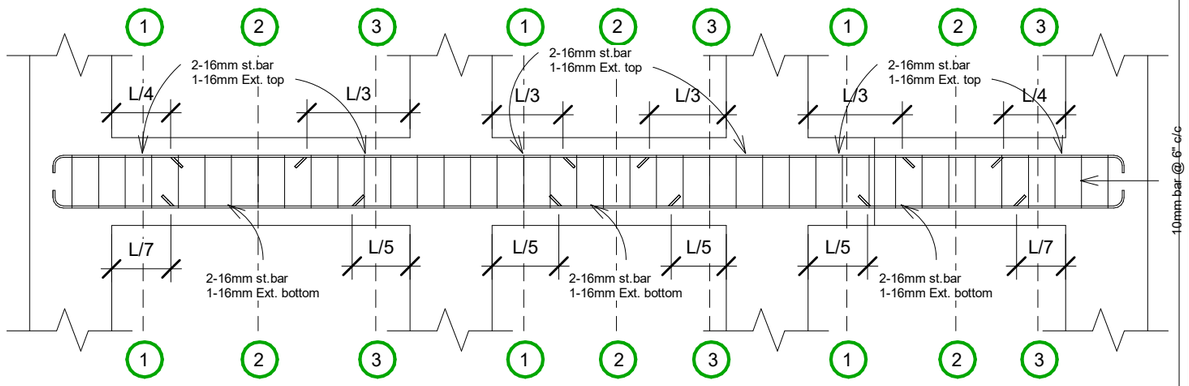


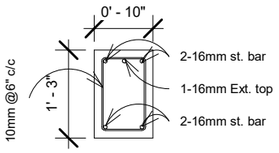
Section 2-2



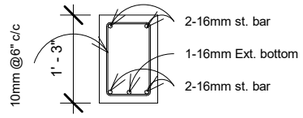
Section 3-3

GB7 @ PL Level

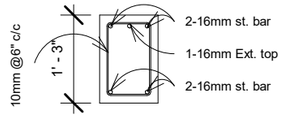




Section 1-1

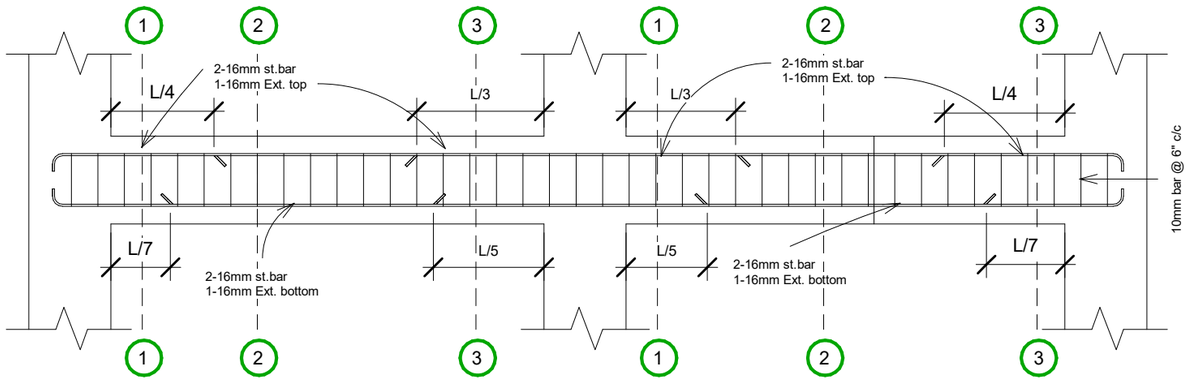


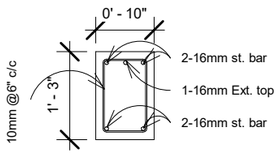
Section 2-2



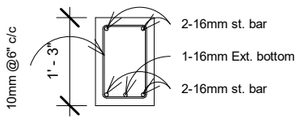
Section 3-3

GB6 @ PL Level

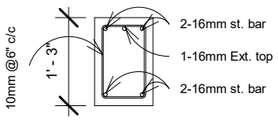




Section 1-1



Section 2-2



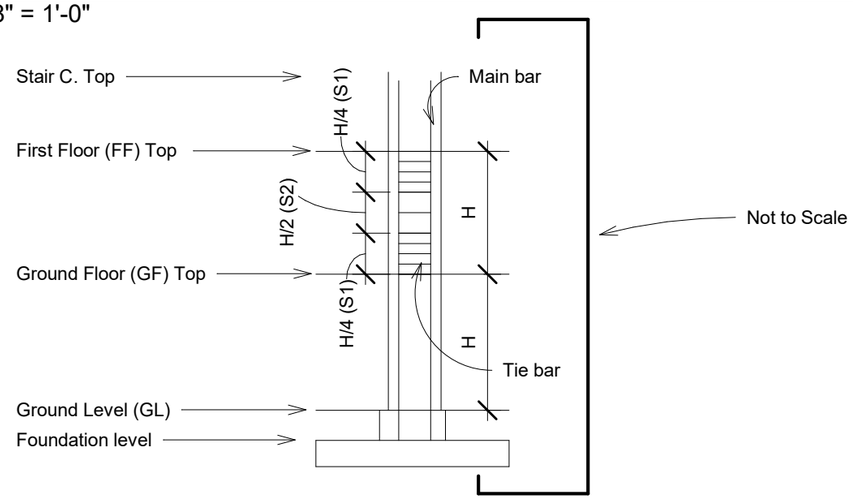
Section 3-3

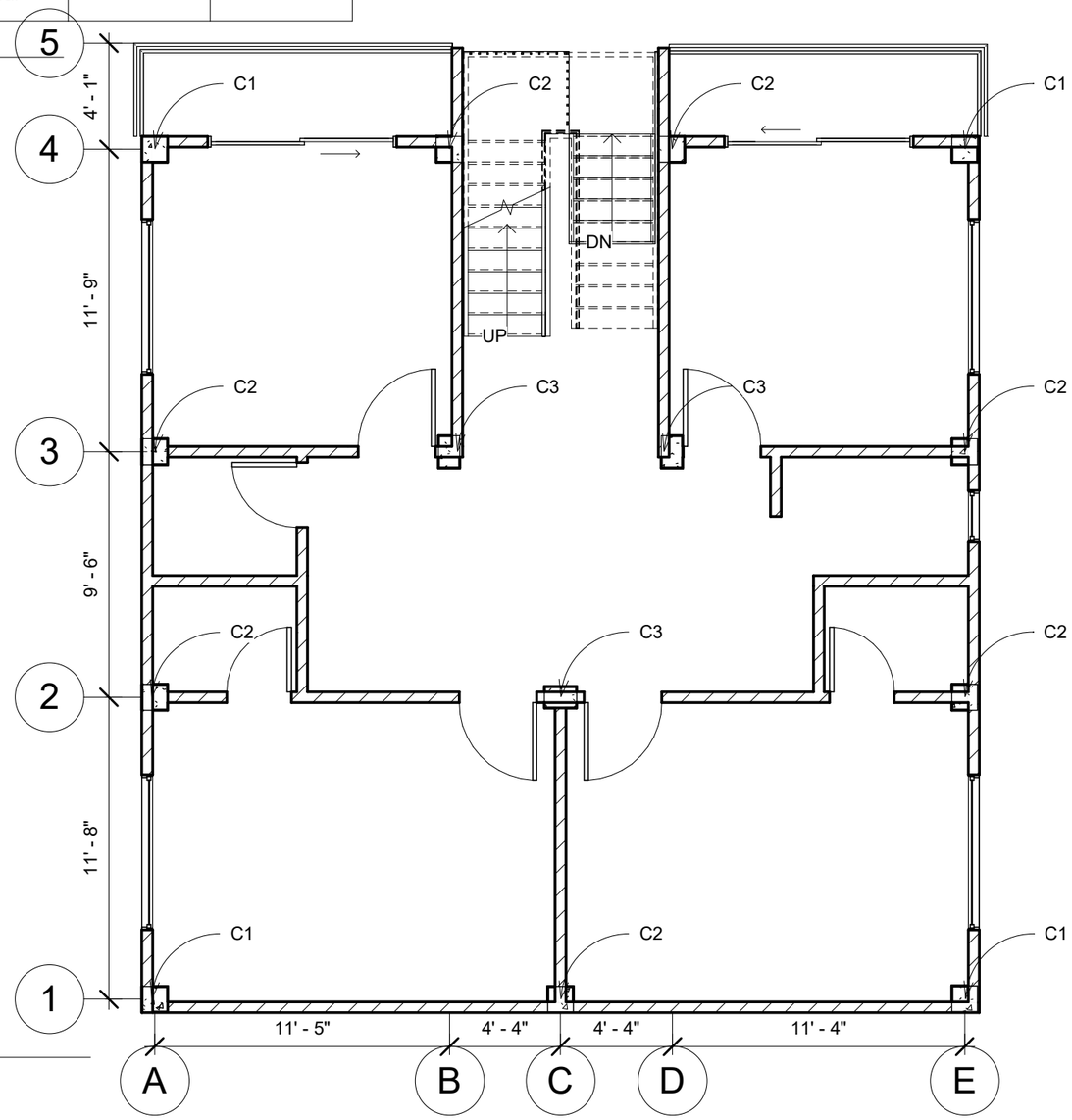
1. Grade Beam Schedule Part 02
* Not to Scale

Project	Authors	Sheet
N2 G+1 Story	Drawn By Shahlan Kabir Designed By Shahlan Kabir Checked By Seikh Arif Istiak	Grade Beam Schedule 02 S04 02/06/26

Tag	Column Size		Column Section From GL to GF Top	Column Section From GF top to 1F Top	Column Section From 1F top to Stair C. Top	10mm Dia Tie Spacing at	
	Below GF	Above GF				Middle Part (H/2) of Column (S2)	Top & Bottom Part (H/4) of Column (S1)
C1	15" x 15"	12" x 12"	4-16mm dia bar	4-16mm dia bar	— — — — —	@ 4" c/c	@ 6" c/c
C2	15" x 15"	12" x 12"	4-16mm dia bar + 2-12mm dia bar	4-16mm dia bar + 2-12mm dia bar	4-16mm dia bar + 2-12mm dia bar	@ 4" c/c	@ 6" c/c
C3	13" x 18"	10" x 15"	6-16mm dia bar + 2-12mm dia bar	6-16mm dia bar + 2-12mm dia bar	6-16mm dia bar + 2-12mm dia bar	@ 4" c/c	@ 6" c/c

1 Column Schedule
3/8" = 1'-0"

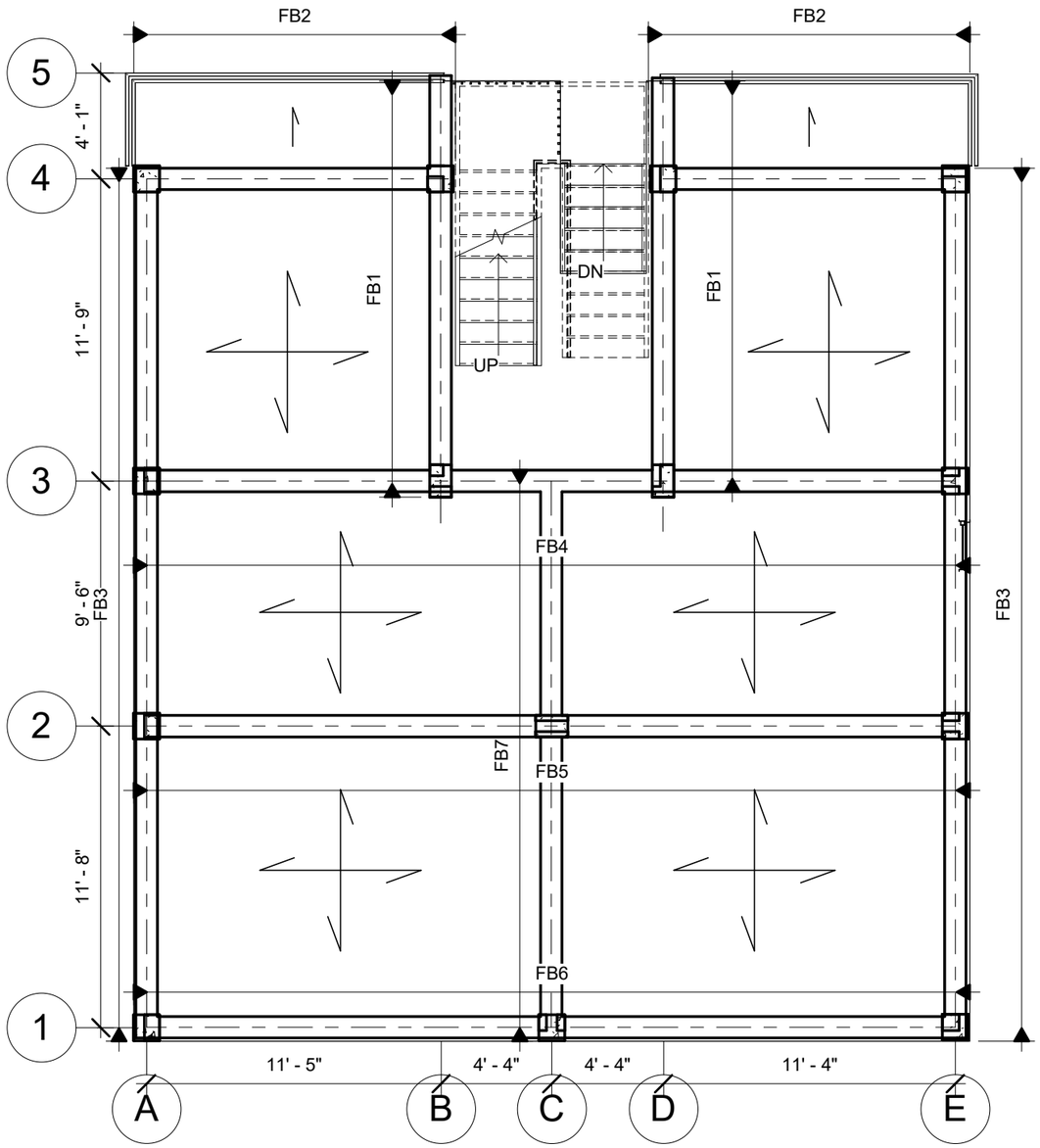




2 Column Layout
3/16" = 1'-0"

1. Column Schedule
2. Column Layout
3. Column Schematic

Project	Authors	Sheet
N2 G+1 Story	Drawn By Shahlan Kabir Designed By Shahlan Kabir Checked By Seikh Arif Istiak	Column Schedule S05 01/28/26



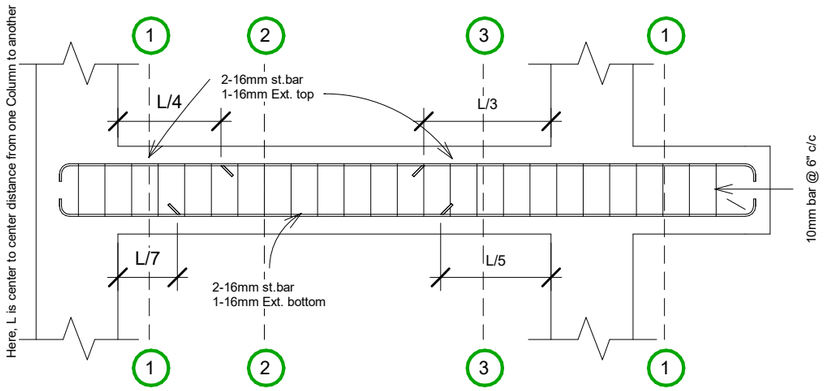
1 Floor Beam and Slab Layout
3/16" = 1'-0"

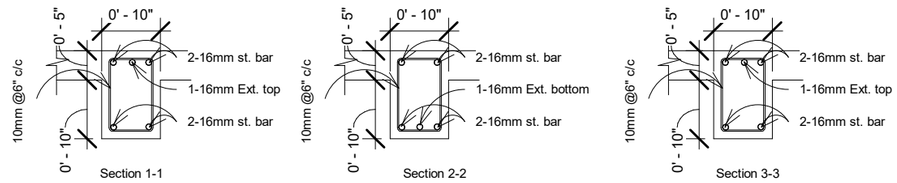
Floor Beam Grid Location		
Tag	**Dim**	**Grid**
FB1	10" x 15"	3B~5B
FB1	10" x 15"	3D~5D
FB2	10" x 15"	5A~5B
FB2	10" x 15"	5D~5E
FB3	10" x 15"	1A~4A
FB3	10" x 15"	1E~4E
FB4	10" x 17"	3A~3E
FB5	10" x 17"	2A~2E
FB6	10" x 15"	1A~1E
FB7	10" x 17"	1C~3C

1. Floor Beam and Slab Layout
2. Floor Beam Grid Location

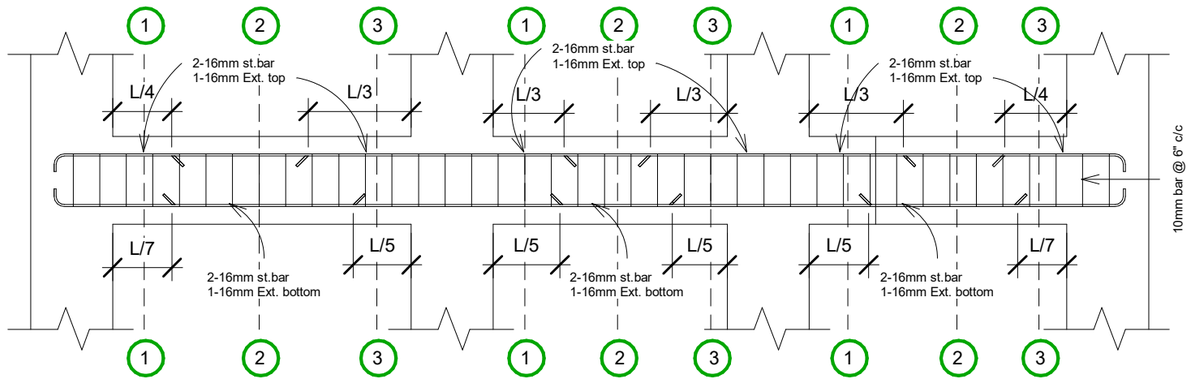
Project	Authors		Sheet
N2 G+1 Story	Drawn By	Shahlan Kabir	Floor Beam
	Designed By	Shahlan Kabir	S06
	Checked By	Seikh Arif Istiak	01/22/26

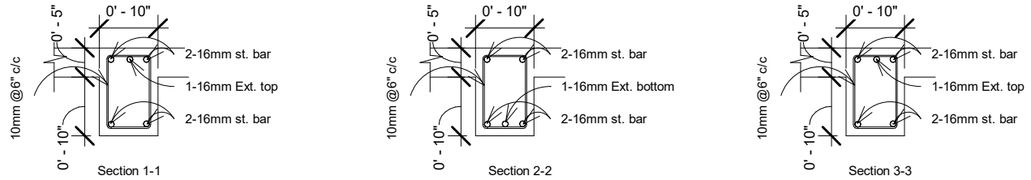
FB1



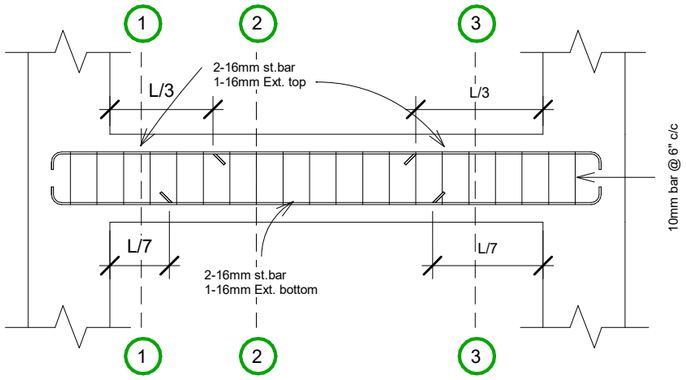


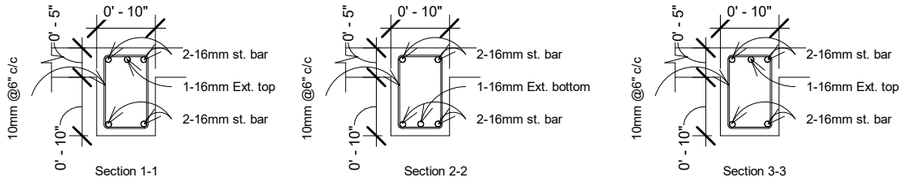
FB3



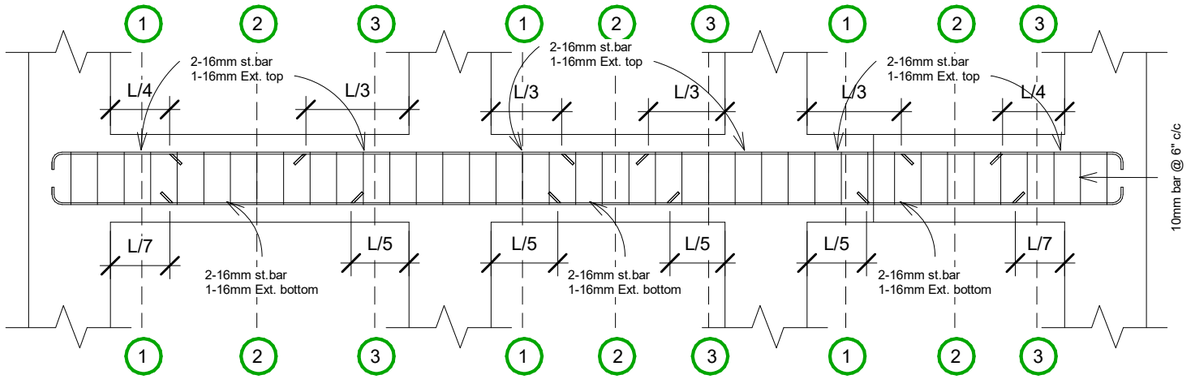


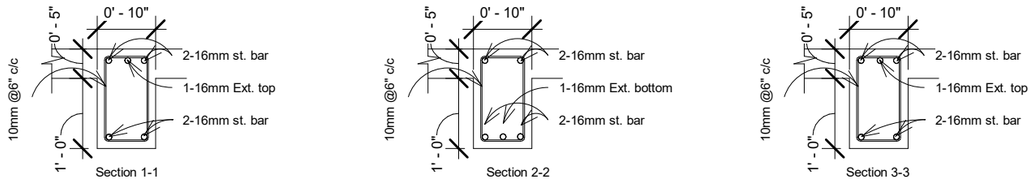
FB2





FB4



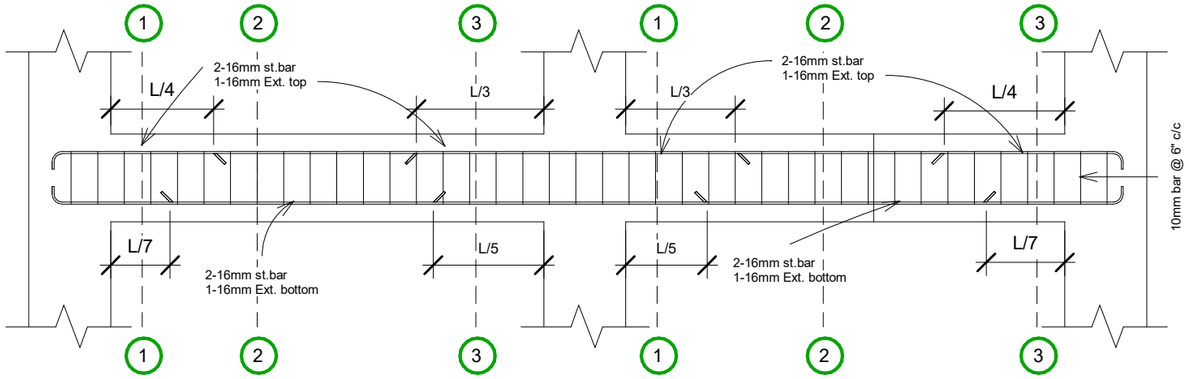


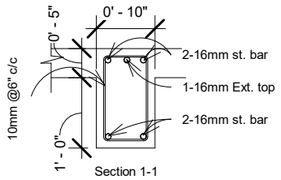
1. Floor Beam Schedule Part 01
* Not to Scale

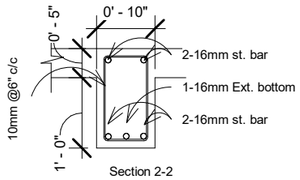
Project	Authors	Sheet
N2 G+1 Story	Drawn By Shahlan Kabir Designed By Shahlan Kabir Checked By Seikh Arif Istiak	Floor Beam Schedule 01 S07 02/06/26

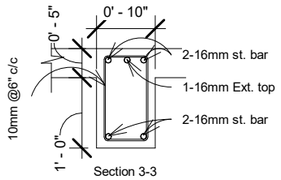
Here, L is center to center distance from one Column to another

FB5

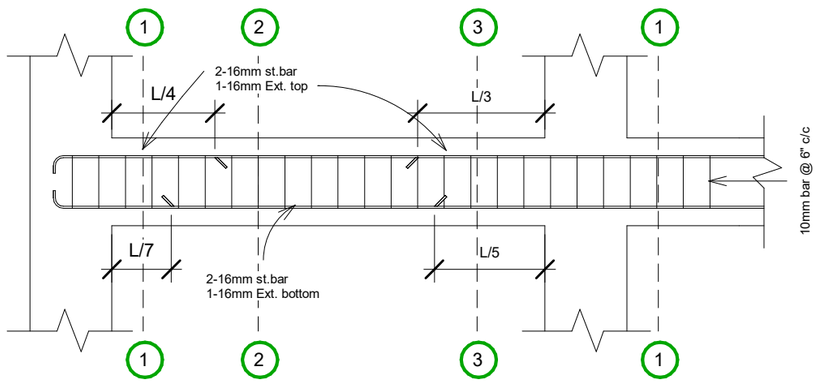


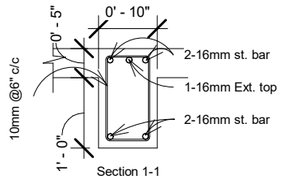


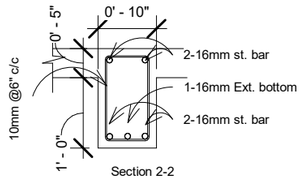


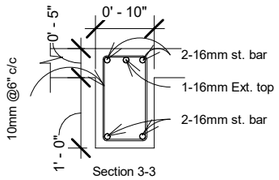


FB7

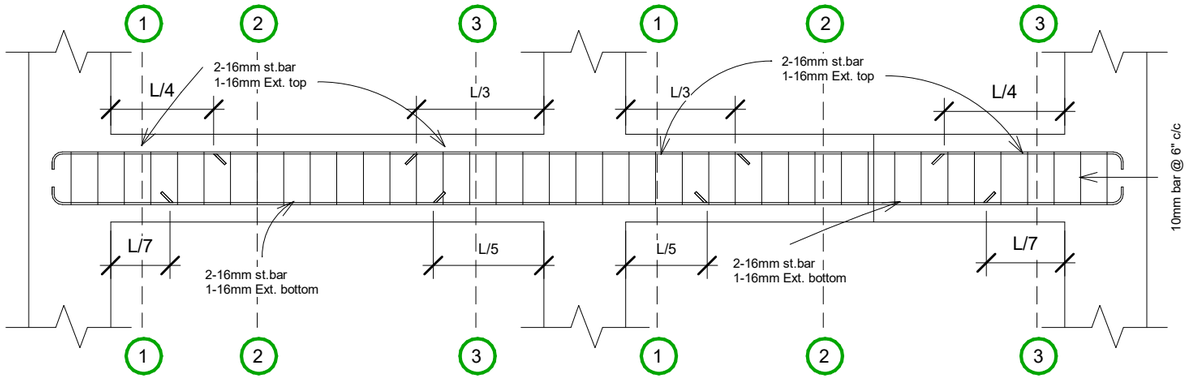


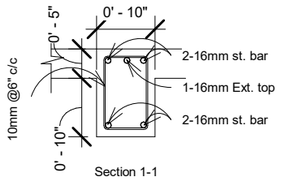


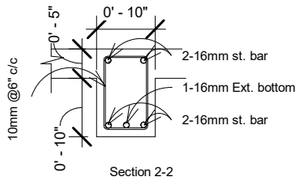


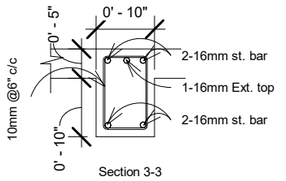


FB6



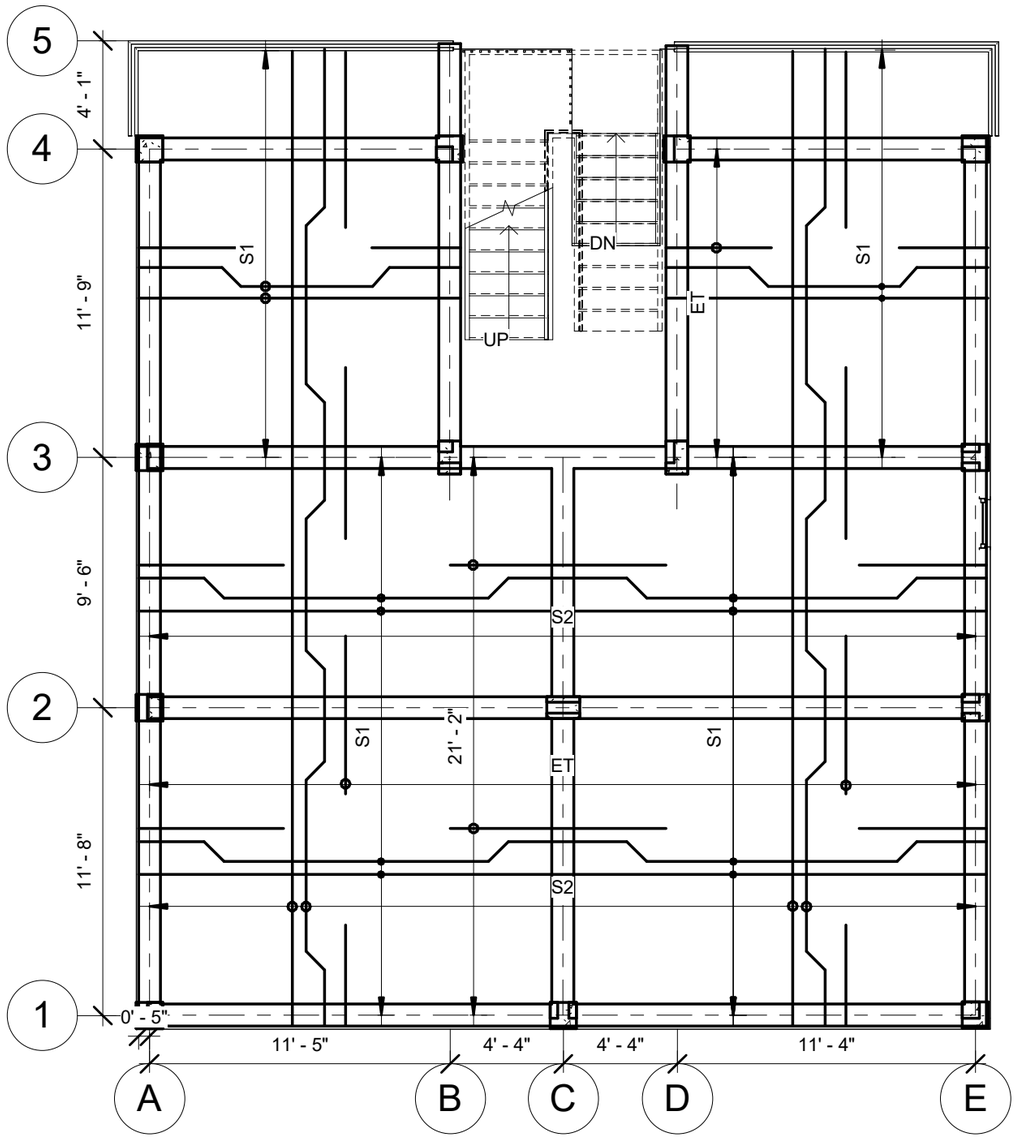






1. Floor Beam Schedule Part 02
* Not to Scale

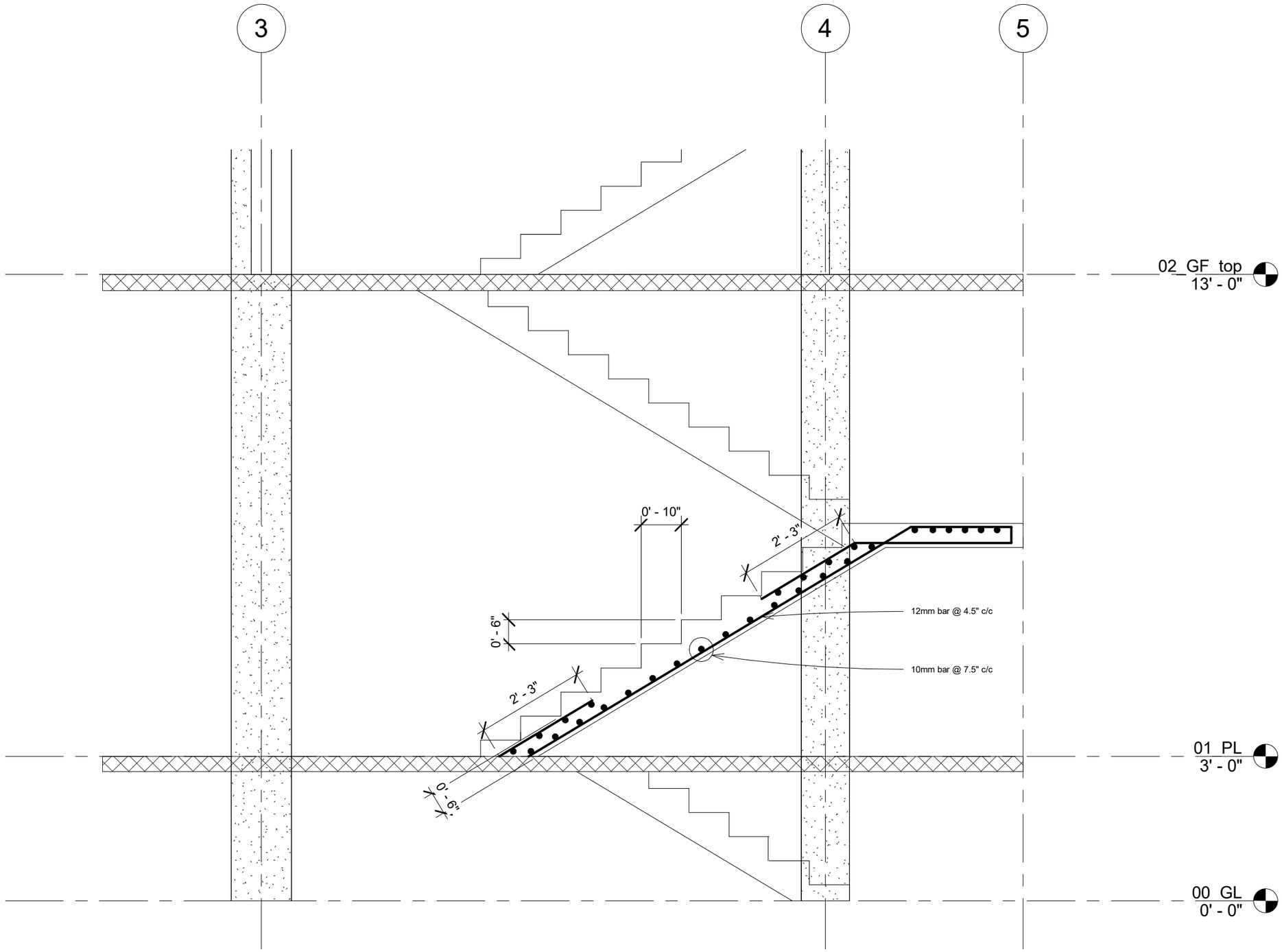
Project	Authors	Sheet
N2 G+1 Story	Drawn By Shahlan Kabir Designed By Shahlan Kabir Checked By Seikh Arif Istiak	Floor Beam Schedule 02 S08 02/06/26



- S1 = Long direction; 10mm bar @ 6"c/c Alt. Ckd.
- S2 = Short direction; 10mm bar @ 5.5"c/c Alt. Ckd.
- ET = Both directin; 1-10mm bar Extra top
- Slab Thicnkess 5"

1 Slab Detailing
3/16" = 1'-0"

Project	Authors		Sheet
N2 G+1 Story	Drawn By	Shahlan Kabir	Slab Detailing
	Designed By	Shahlan Kabir	S09
	Checked By	Seikh Arif Istiak	02/07/26



Note:
- Waist Slab Thickness 6"
- Thread 19 Nos @ 10" c/c
- Raiser 20 Nos @ 6" c/c

1 Section 1 - Callout 1
1/2" = 1'-0"

Project	Authors		Sheet
N2 G+1 Story	Drawn By	Shahlan Kabir	Stair Detailings
	Designed By	Shahlan Kabir	S10
	Checked By	Seikh Arif Istiak	02/07/26